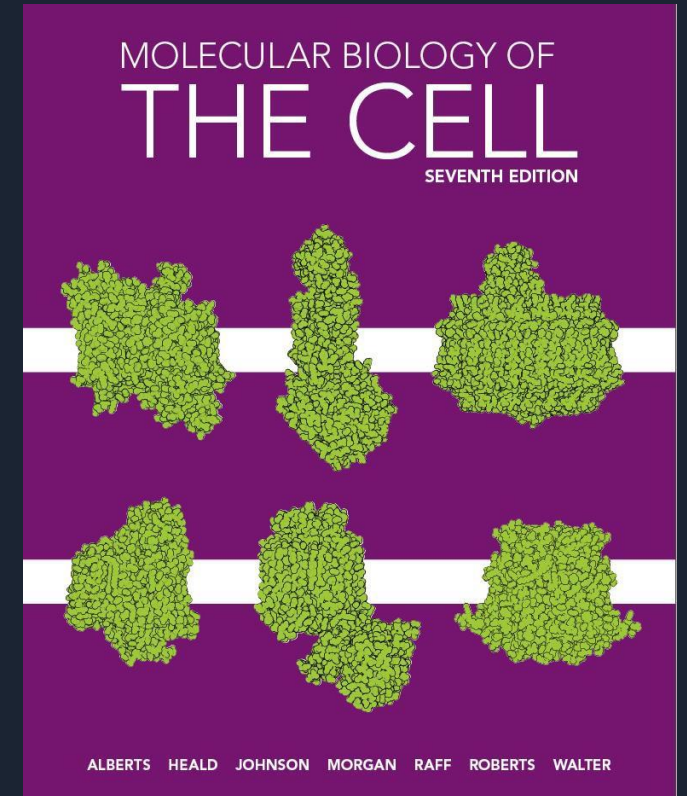
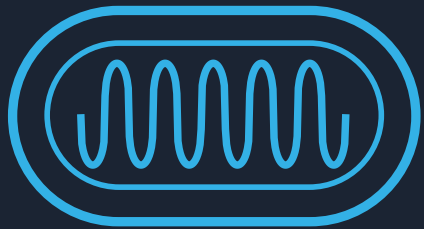


MOLECULAR MEDICINE • MSc LECTURE

Mitochondria

Energy conversion in health and disease

Alessandro Iuliano, PhD
2026



Reading: Chapter 14, pp. 811–842

Lecture roadmap

01

Why mitochondria matter

Energy, metabolism, cell fate

02

Origin & structure

Endosymbiosis, dynamics, two membranes

03

Chemiosmosis

Coupling electron flow to ATP via H^+

04

NADH

From fuel to high-energy electrons

05

Proton-motive force

$\Delta V + \Delta pH \approx 180 \text{ mV}$

06

Why protons?

Fast H^+ transfer through water

07

Respiratory chain

Complexes I–IV and their carriers

08

ATP synthase

Rotary motor, reversibility, transport

09

Case: muscle

Fibre types, mitochondria, Ca^{2+}

10

Disease

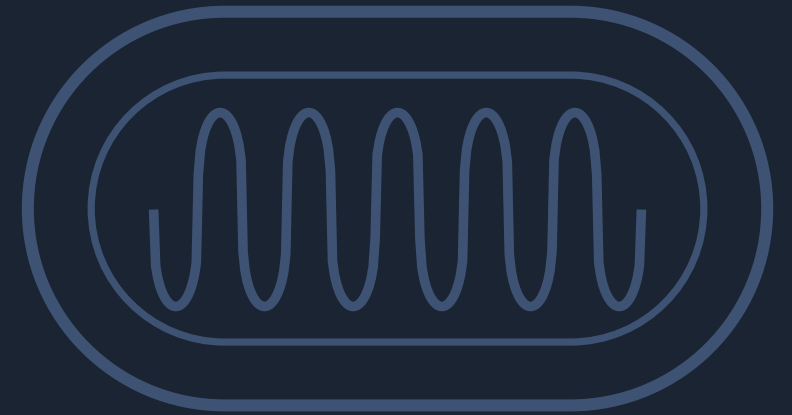
mtDNA vs nuclear defects; donation

Reading: Alberts et al., *Molecular Biology of the Cell*, Chapter 14 (pp. 811–842)

01

Why are mitochondria so important for eukaryotes?

Energy · metabolism · cell fate



Mitochondria turn food into usable energy...and much more

~30

ATP per glucose when it is fully oxidised in mitochondria

2

ATP per glucose from glycolysis alone



Energy conversion

Oxidative phosphorylation supplies most of the ATP in aerobic cells.



Metabolic hub

Citric acid cycle, fatty-acid β -oxidation, and precursors for amino acids, heme and lipids.



Signalling & cell fate

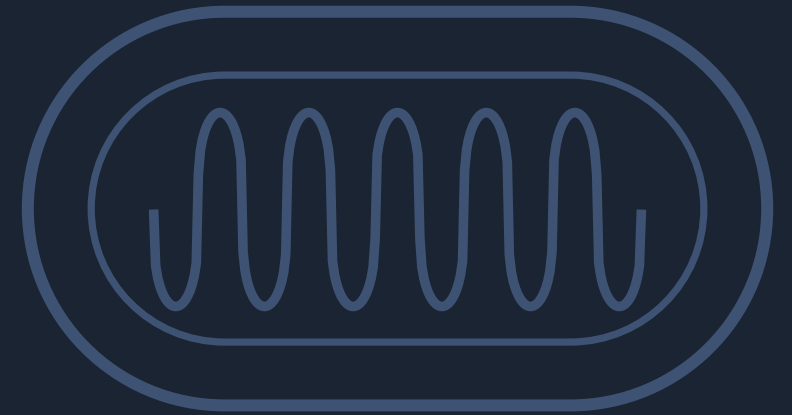
Ca^{2+} buffering, ROS signalling, and apoptosis via cytochrome c release.

Not only ATP: cells lacking mtDNA (ρ^0 cells) survive on glycolysis when given pyruvate and uridine — yet the organelle itself stays essential, e.g. for iron–sulfur cluster assembly.

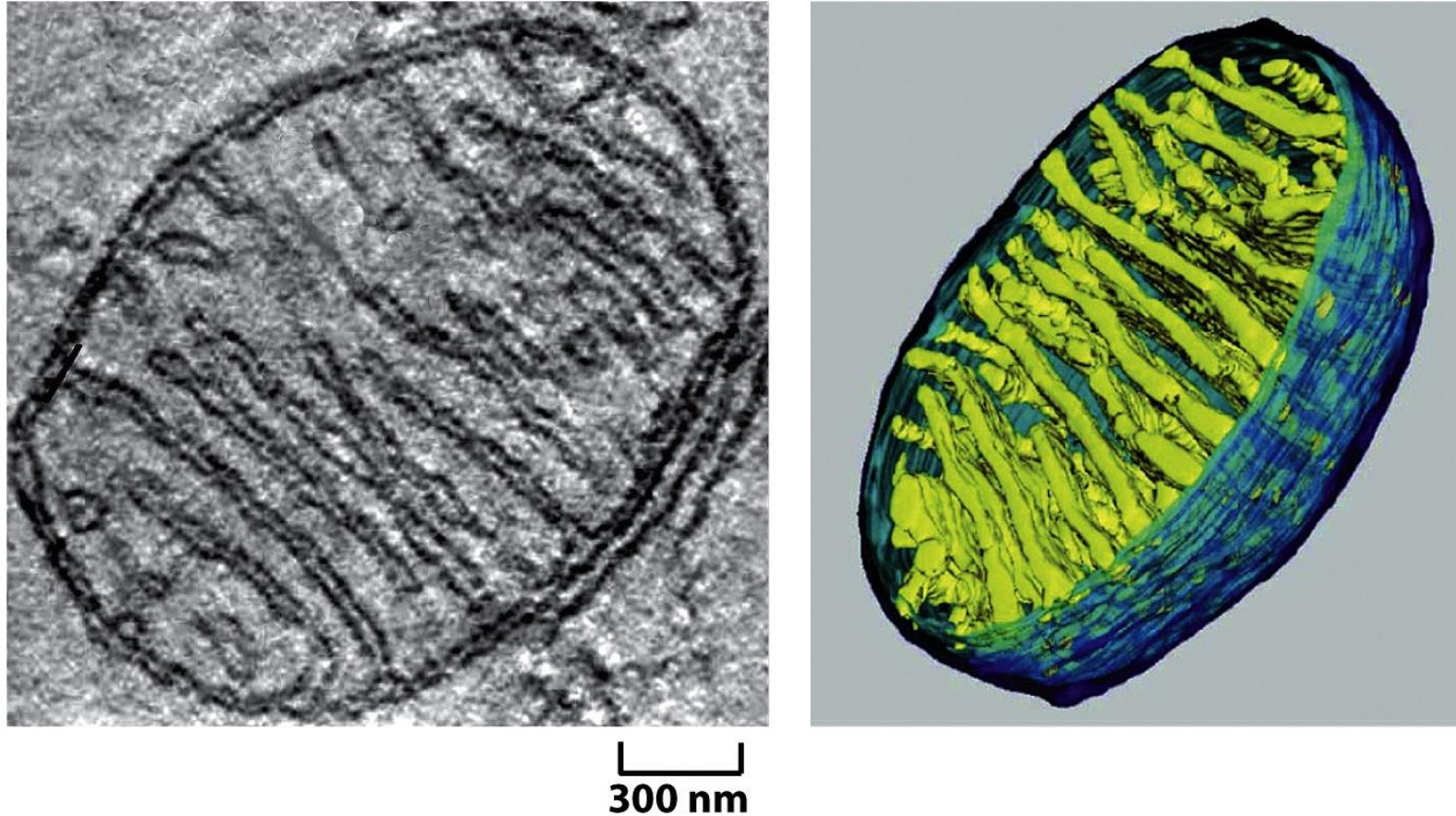
02

How did eukaryotes acquire mitochondria and chloroplasts?

Endosymbiosis · dynamics · two membranes

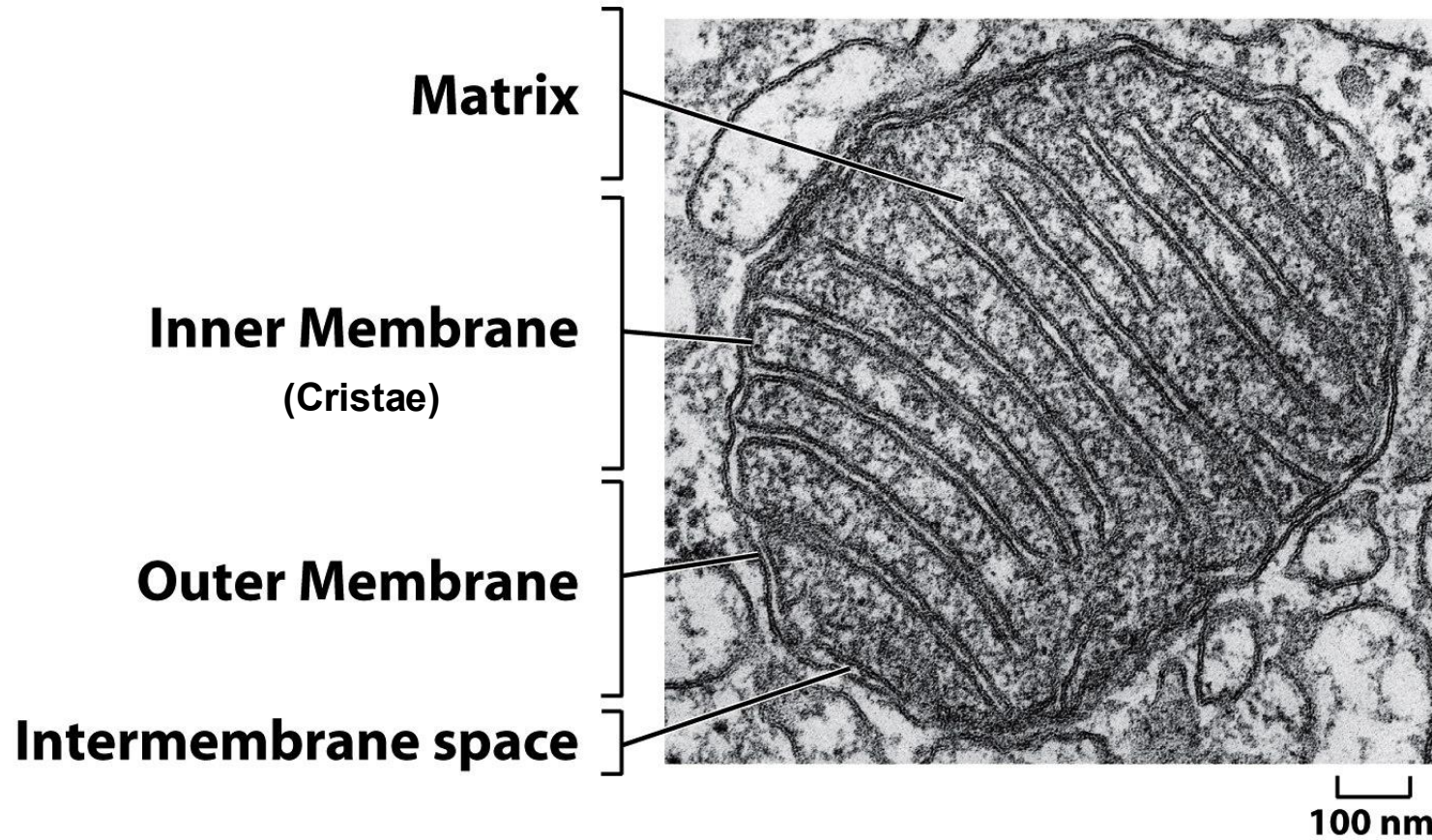


Electron tomography reveals the 3D architecture of the inner membrane



Left: a single tomographic slice through a mitochondrion. Right: 3D reconstruction — the inner membrane folds into numerous cristae (yellow) inside the outer envelope (blue). Scale bar 300 nm.

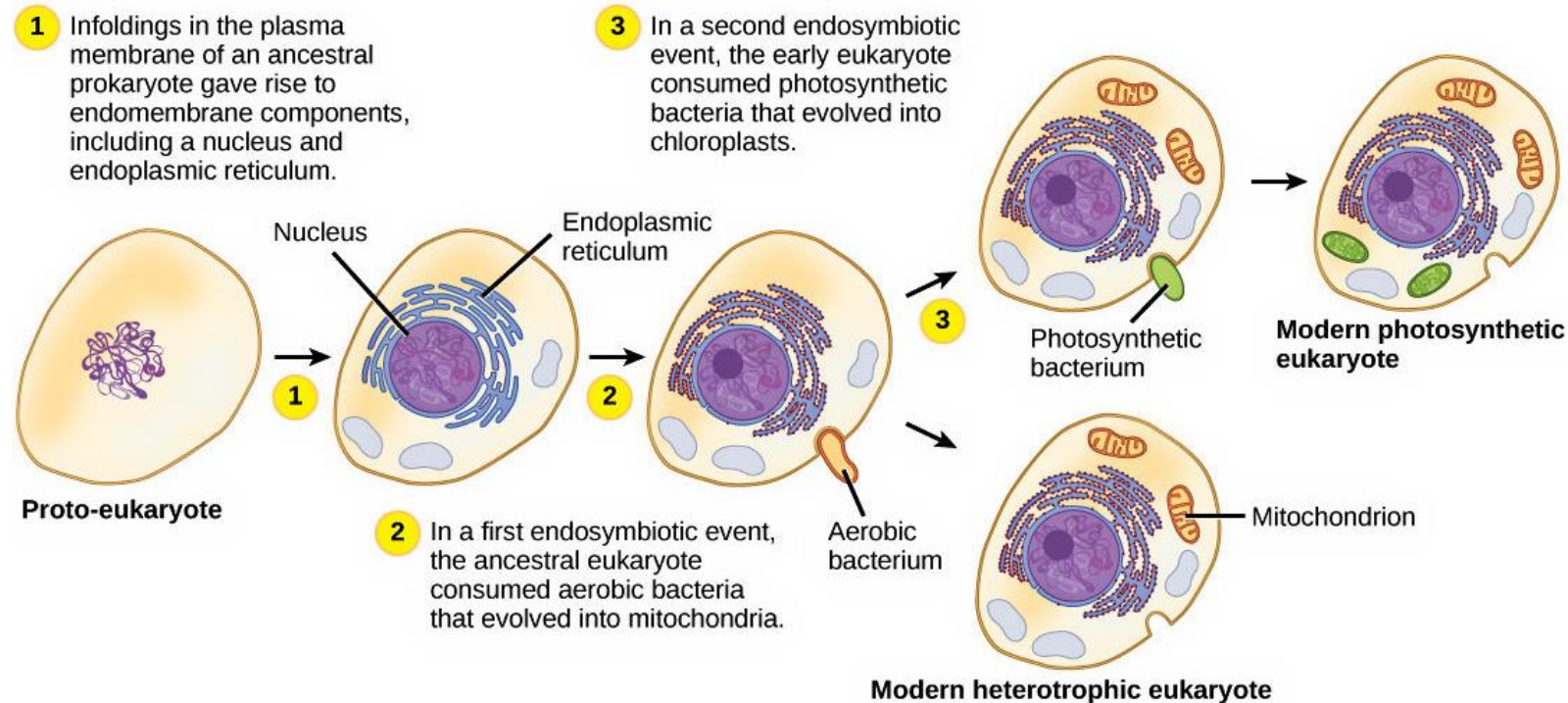
Two membranes create four compartments



KEY POINTS

- **Outer membrane** — permeable to small molecules through porins.
- **Intermembrane space** — where pumped H^+ accumulate.
- **Inner membrane (cristae)** — ion-impermeable; holds the electron-transport chain and ATP synthase.
- **Matrix** — citric acid cycle, fatty-acid oxidation, mtDNA and mitochondrial ribosomes.

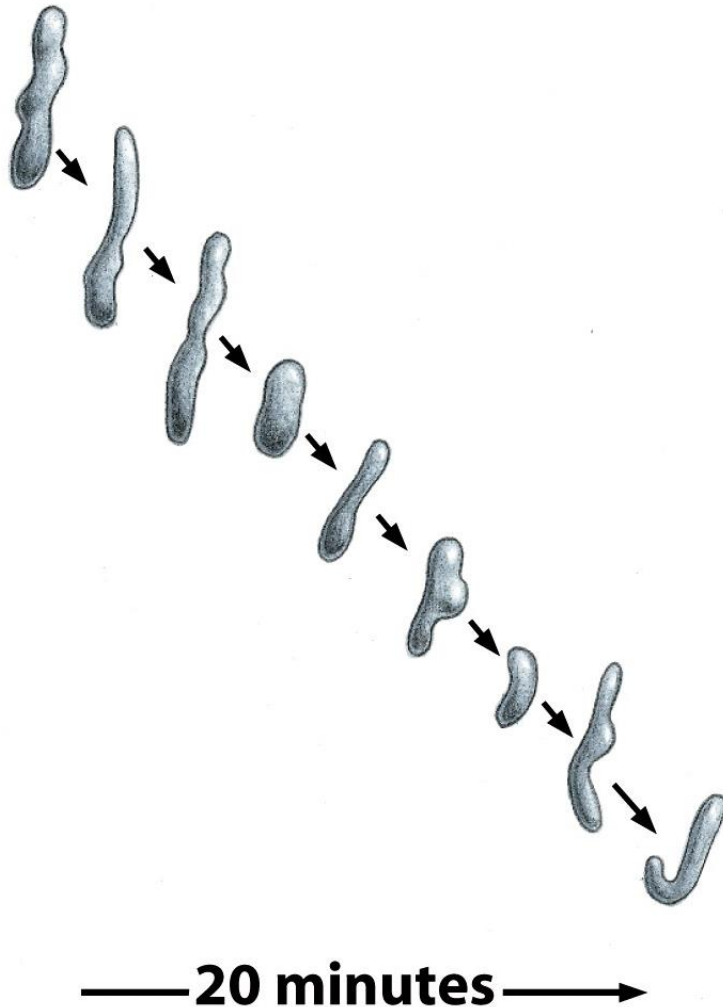
Endosymbiosis: mitochondria descend from an engulfed aerobic bacterium



EVIDENCE FOR A BACTERIAL ORIGIN

- **Two membranes** — the inner one derives from the bacterial plasma membrane.
- **Own circular genome** (mtDNA) and bacterial-type ribosomes — sensitive to some antibiotics.
- **Division by fission** — never assembled de novo.
- **Closest living relatives** — α -proteobacteria (chloroplasts: cyanobacteria).

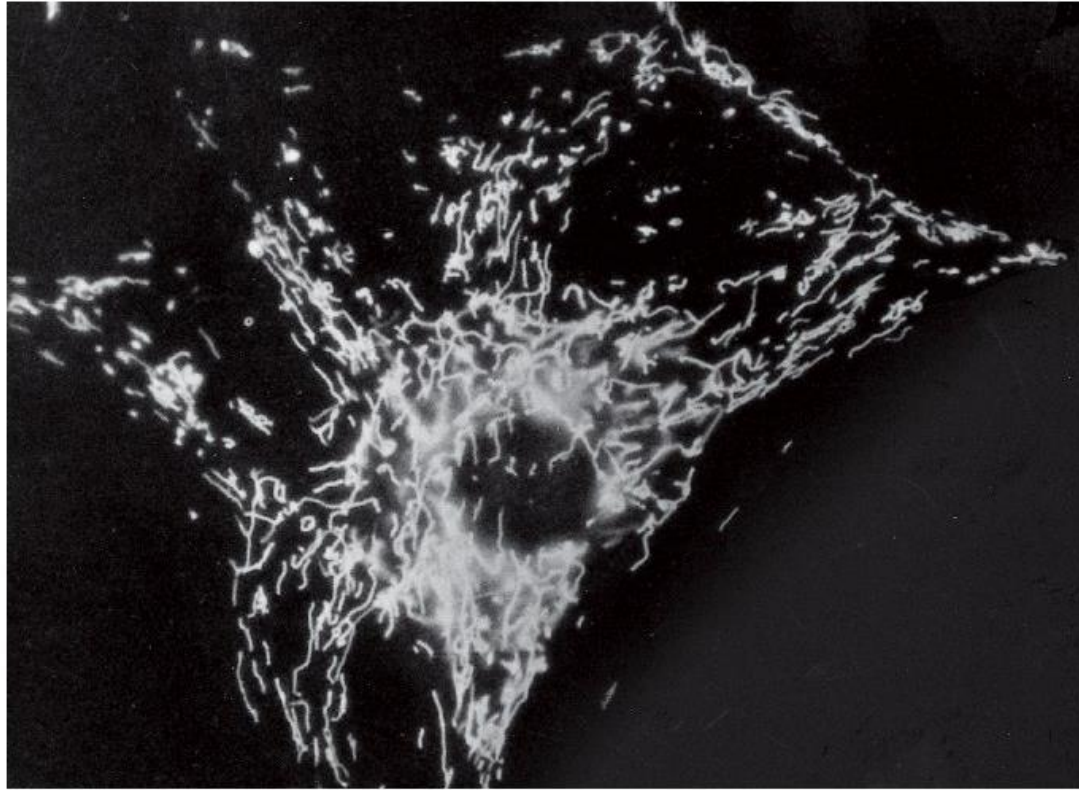
Mitochondria are dynamic: they continuously change shape, fuse and divide



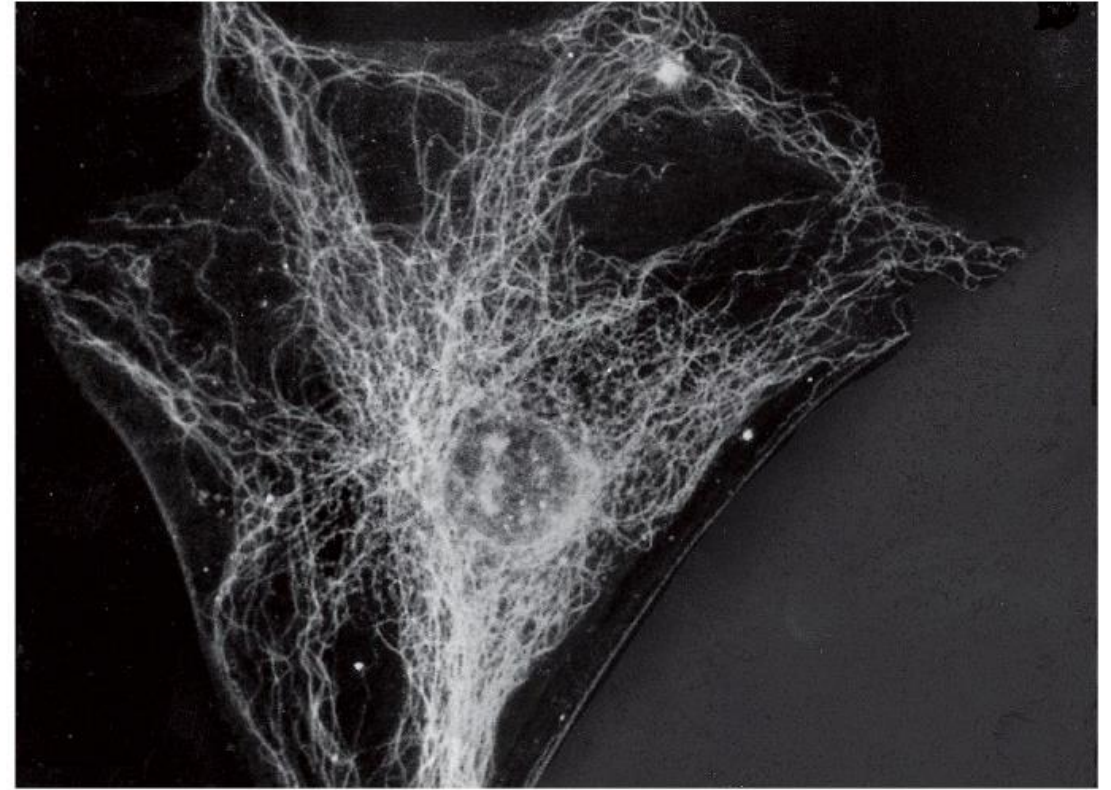
KEY POINTS

- **Figure** — one mitochondrion followed in a living cell; its shape changes within 20 minutes.
- **Fusion** — MFN1/MFN2 (outer membrane) and OPA1 (inner membrane).
- **Fission** — DRP1 (gene *DNM1L*), recruited by MFF.
- **Why it matters** — content mixing, quality control (mitophagy) and distribution; *OPA1*, *MFN2* and *DNM1L* mutations cause disease (section 10).

Mitochondria are distributed along microtubules



(A)

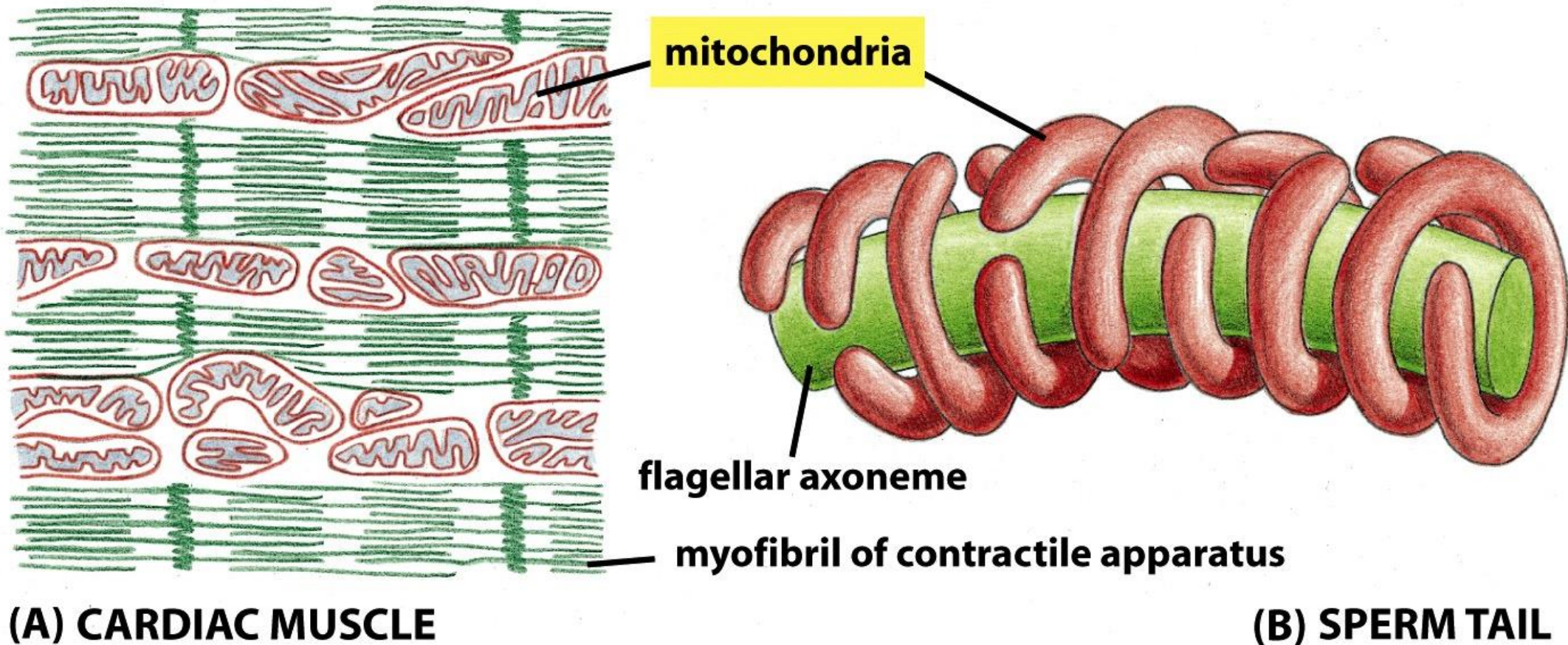


(B)

10 μm

Fluorescence micrographs of the same cultured cell: (A) elongated mitochondria; (B) microtubules. Mitochondria align with, and are moved along, the microtubule network. Scale bar 10 μm .

Mitochondria are packed where ATP is consumed



(A) Cardiac muscle: mitochondria lie between the myofibrils of the contractile apparatus. (B) Sperm tail: mitochondria wrap around the flagellar axoneme that drives swimming.

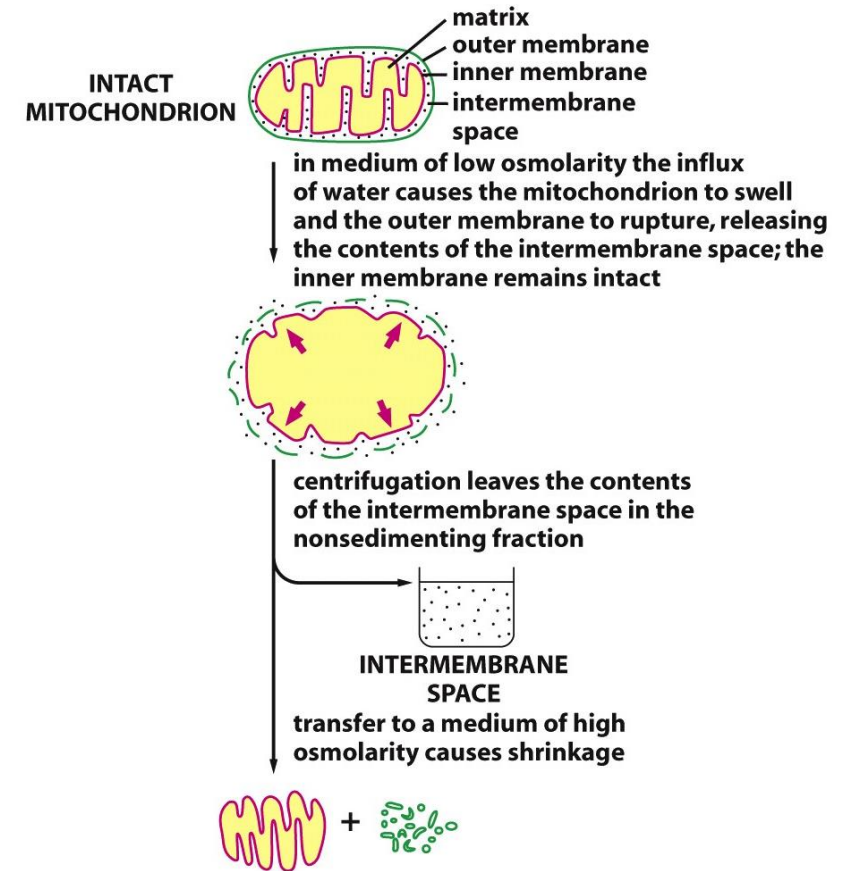
Two membranes with very different permeability — they can be separated

Outer membrane

- **Porins** (VDAC) form large aqueous channels
- Permeable to molecules up to ~5,000 Da (ions, ATP, pyruvate)
- Intermembrane space $\approx$ cytosol for small molecules

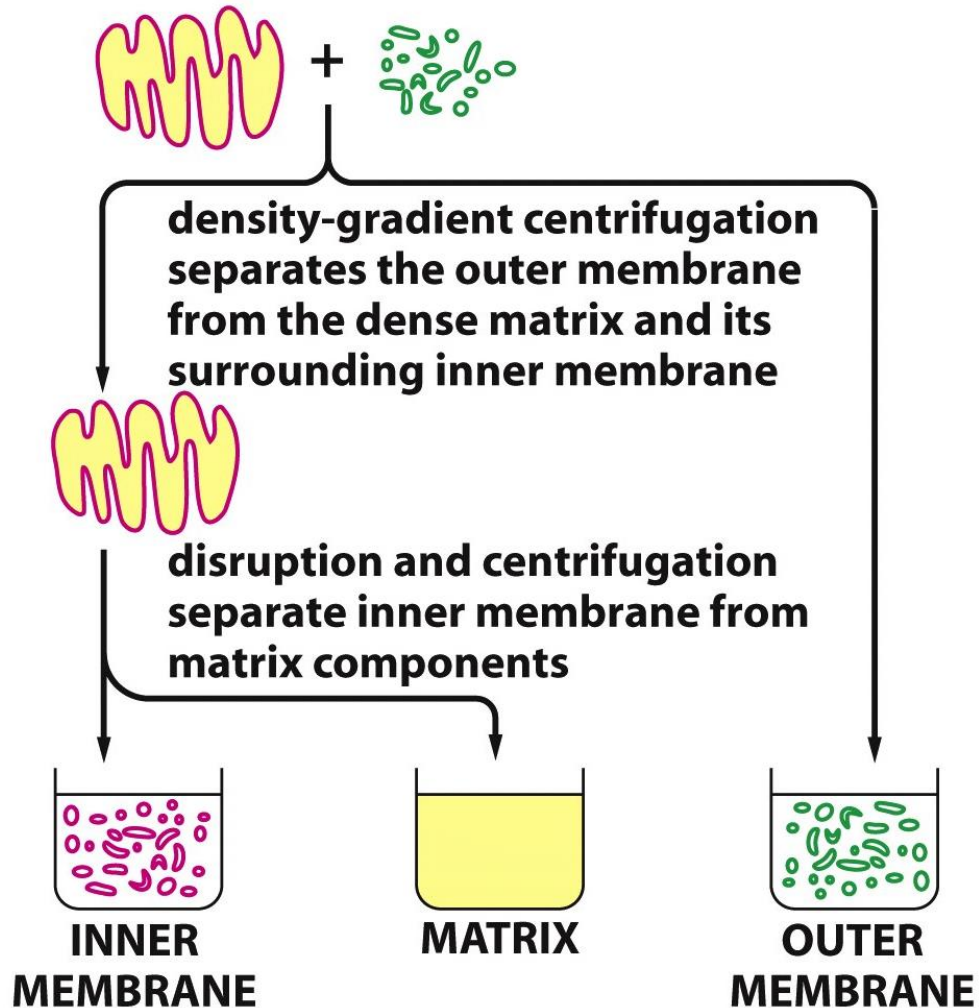
Inner membrane

- Rich in **cardiolipin**; densely packed with proteins
- Impermeable to ions and most small molecules $\rightarrow$ can hold an H^+ gradient
- Traffic only via selective carriers (SLC25 family) and channels



Fractionation, step 1: in a hypotonic medium the matrix swells and ruptures the outer membrane, releasing the intermembrane-space contents.

Centrifugation then isolates outer membrane, inner membrane and matrix



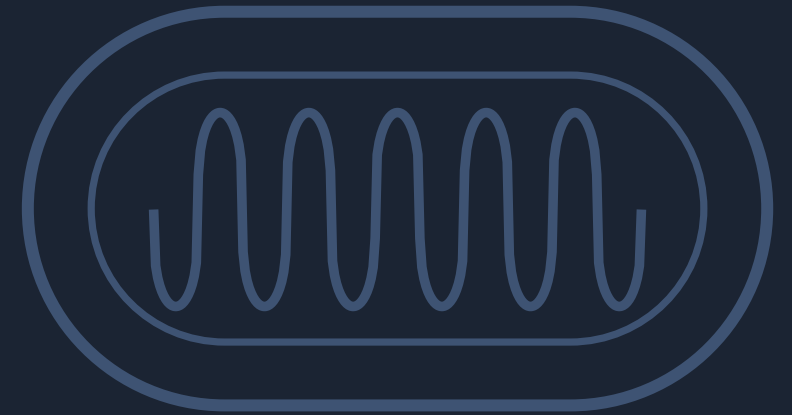
KEY POINTS

- After swelling, density-gradient centrifugation separates the light outer membrane from the dense matrix still enclosed by the inner membrane.
- Disrupting that fraction and centrifuging again separates inner membrane from soluble matrix components.
- Each compartment can then be analysed for its own proteins and enzymes.

03

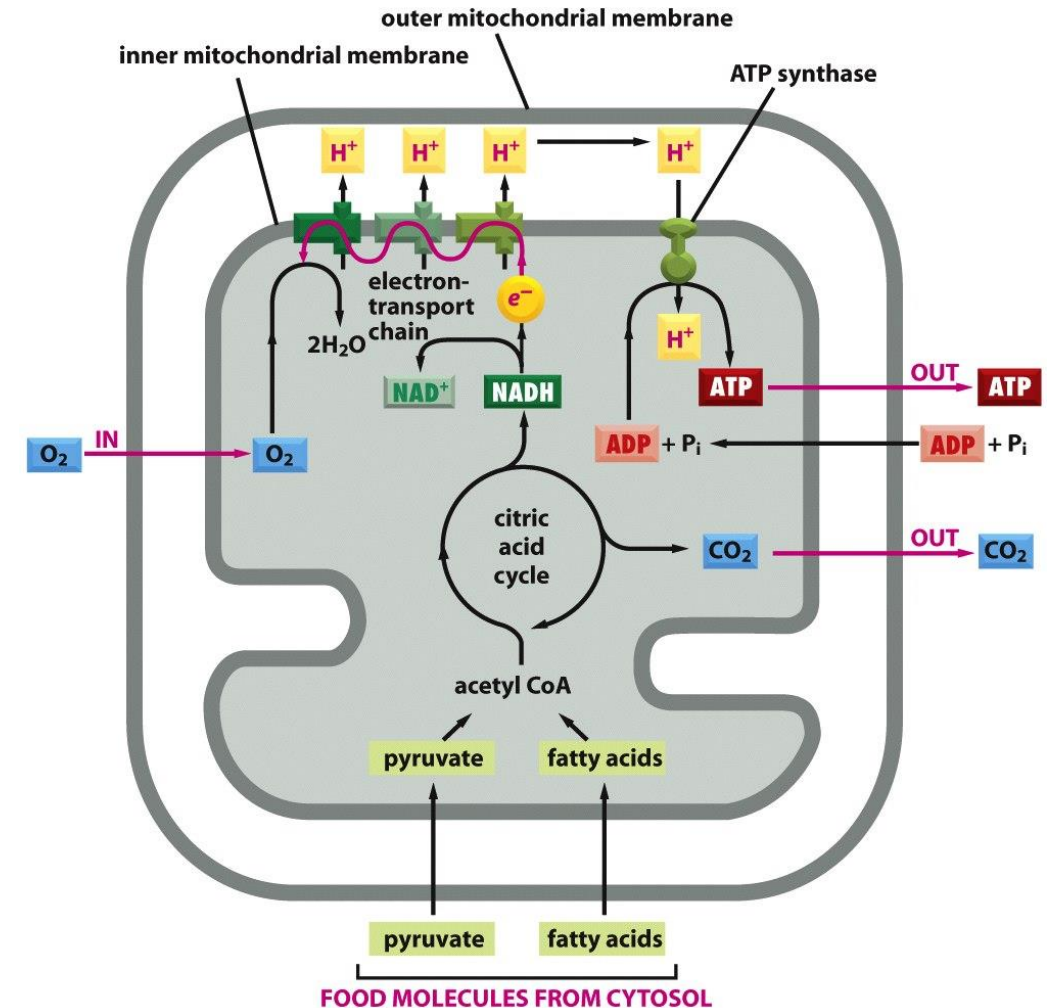
What basic mechanism do mitochondria use to deliver energy to the cell?

Chemiosmosis, and the architecture that makes it possible

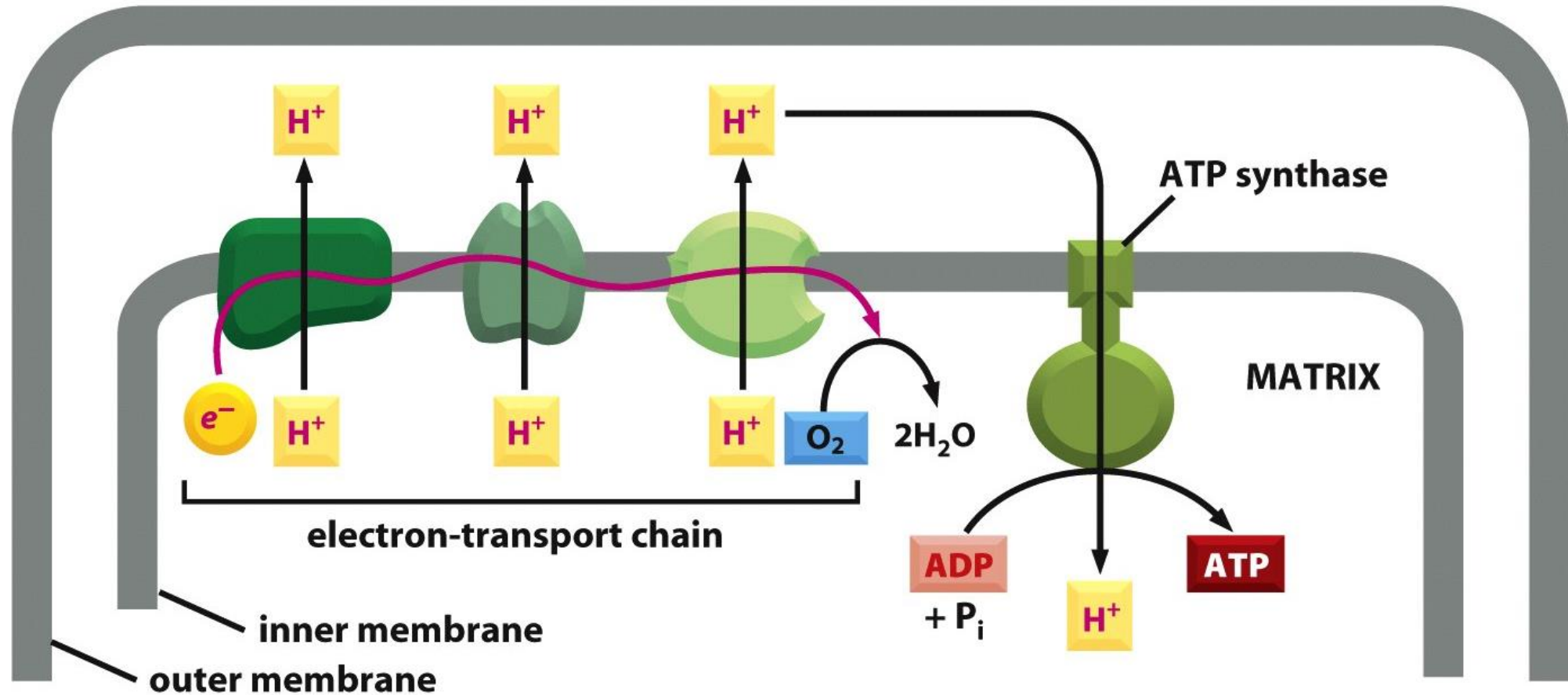


Overview: fuel is oxidised in the matrix to produce high-energy electrons

- 1 **Fuel** — glucose → pyruvate (glycolysis, cytosol); fats → fatty acids
- 2 **Import** across the inner membrane into the matrix
- 3 **Acetyl CoA** — from pyruvate (pyruvate dehydrogenase) or fatty acids (β-oxidation)
- 4 **Citric acid cycle** oxidises the acetyl groups to CO₂
- 5 **High-energy electrons** are captured by NADH and FADH₂ → electron-transport chain

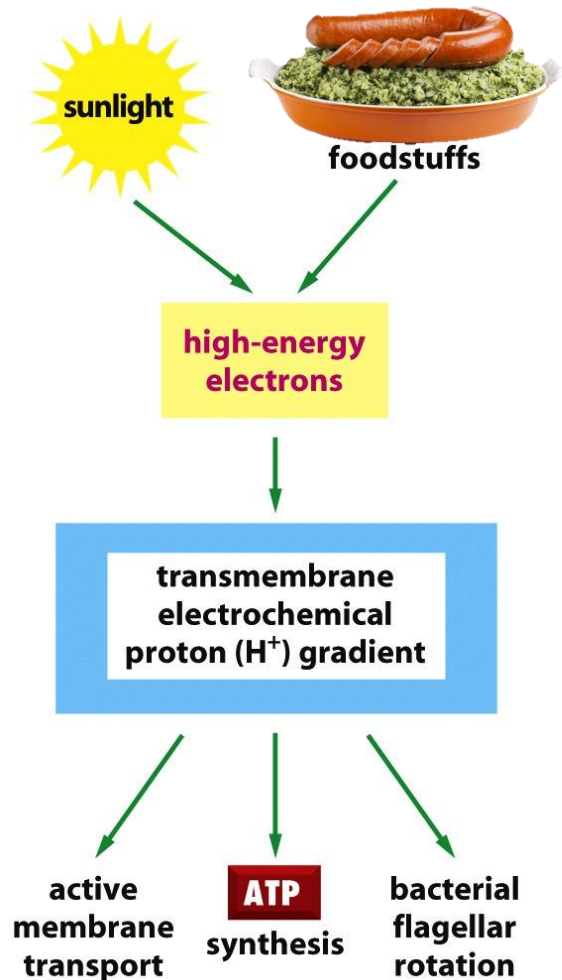


The electron-transport chain builds an H^+ gradient that ATP synthase uses



NADH donates electrons to the chain in the inner membrane. H^+ are pumped into the intermembrane space and flow back through ATP synthase, making ATP from $ADP + P_i$. O_2 accepts the electrons, forming H_2O .

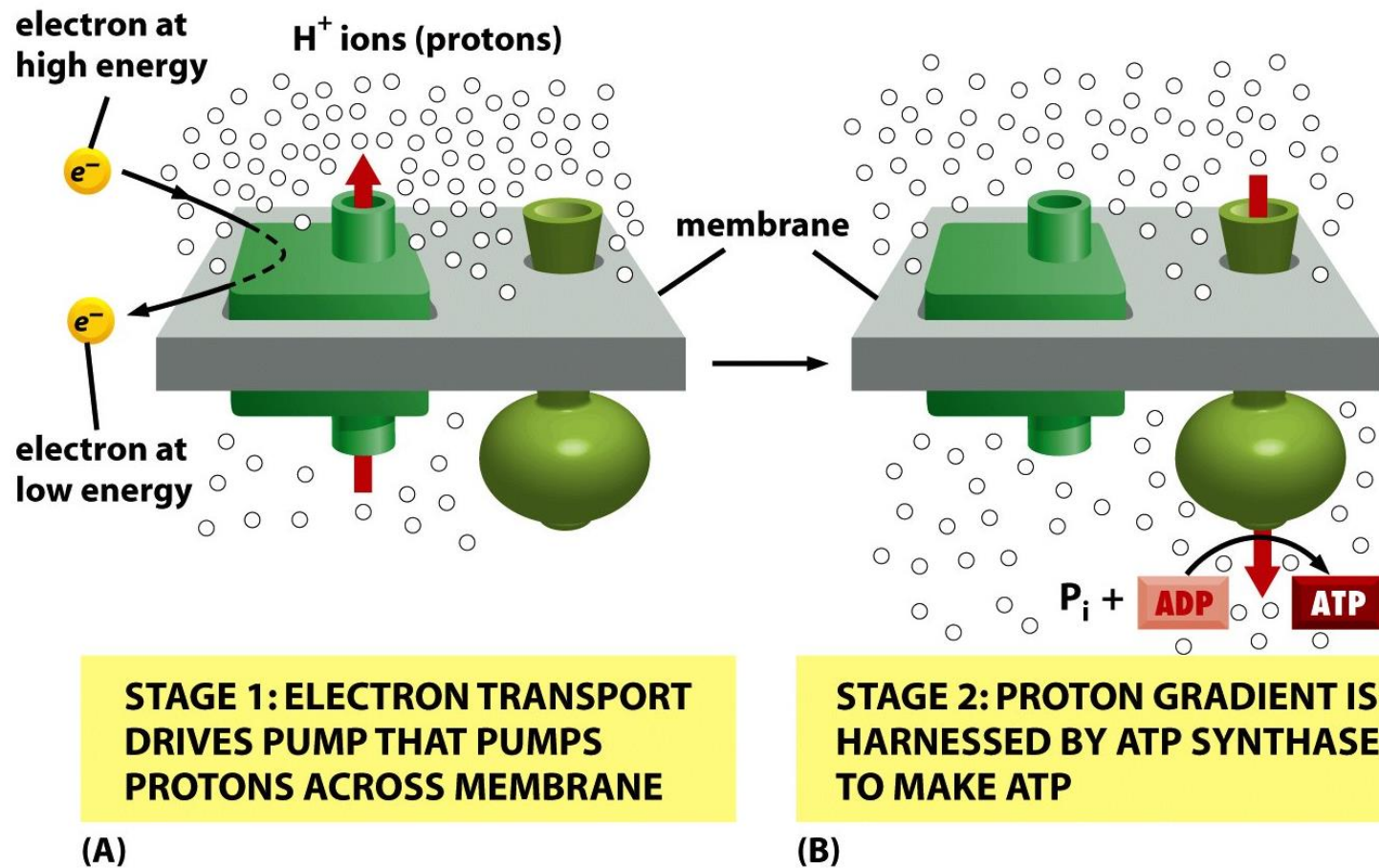
Energy from food or sunlight is stored transiently as an H^+ gradient



KEY POINTS

- **Input** — high-energy electrons from oxidised foodstuffs (mitochondria) or from light-excited chlorophyll (chloroplasts).
- **Common currency** — the transmembrane electrochemical H^+ gradient.
- **Outputs** — ATP synthesis, active membrane transport and, in bacteria, flagellar rotation.

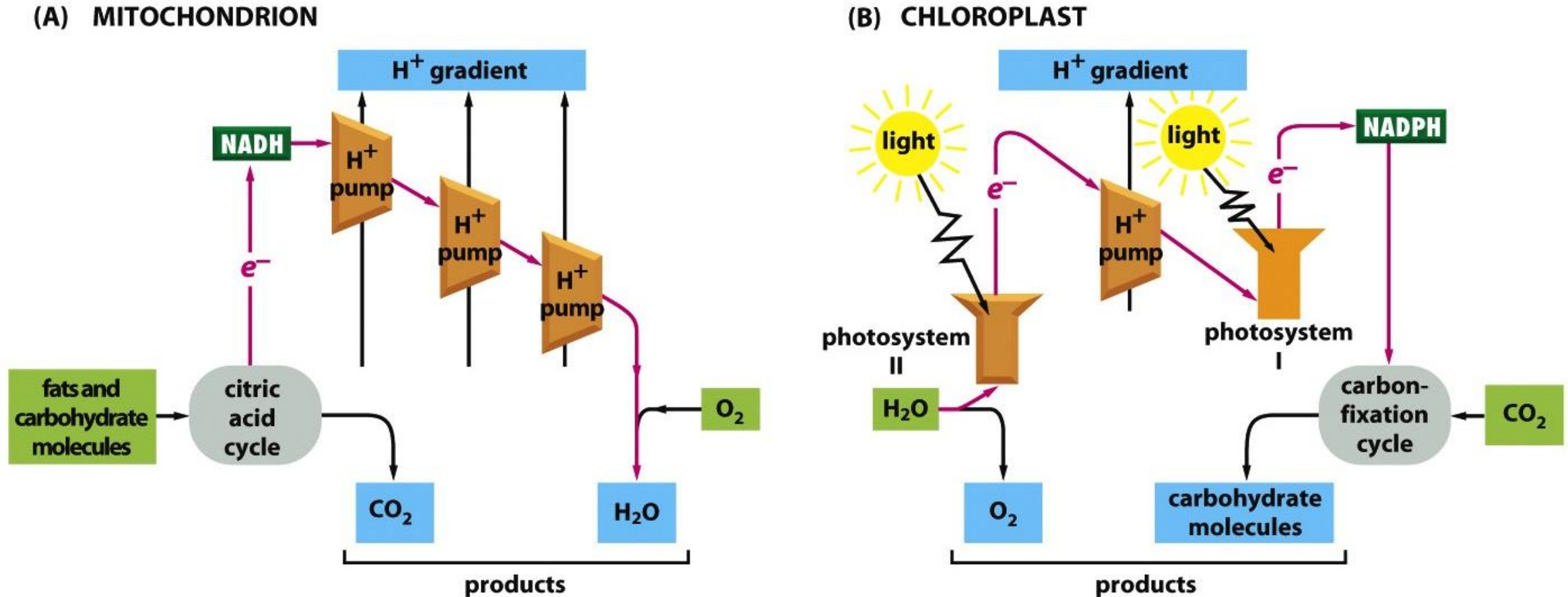
Chemiosmosis couples electron transport to ATP synthesis via an H^+ gradient



KEY POINTS

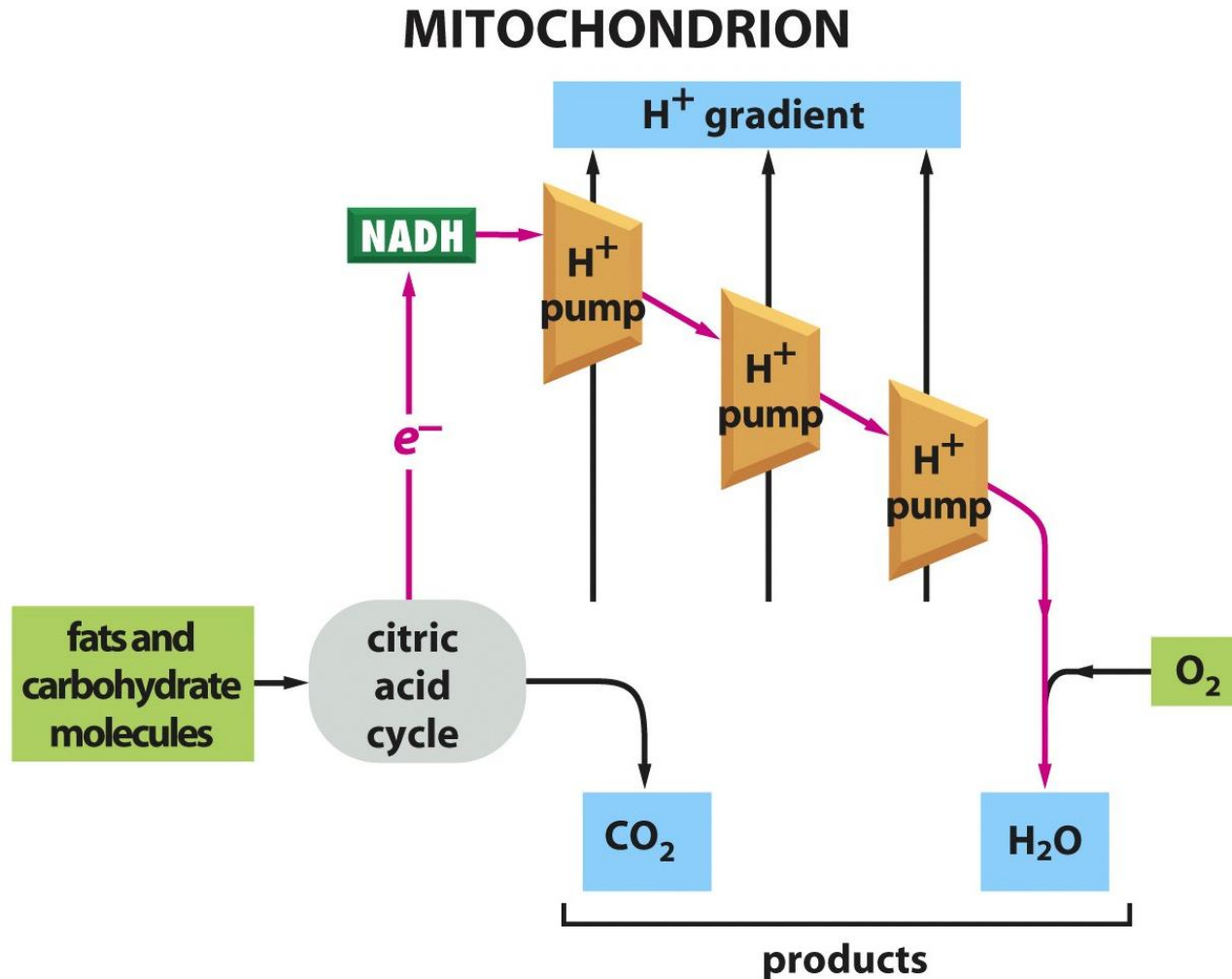
- **Stage 1** — electrons pass from high to low energy through membrane proteins that use the released energy to pump H^+ across the membrane.
- **Stage 2** — H^+ flow back down their gradient through ATP synthase, which converts that energy into ATP.
- **The name** — *chemi* = the redox chemistry; *osmosis* = the ion transport across the membrane.

Mitochondria and chloroplasts run the same chemiosmotic logic



(A) Mitochondrion: electrons from food reduce O_2 to H_2O , and CO_2 is released. (B) Chloroplast: light pulls electrons from H_2O (releasing O_2) to make NADPH, which fixes CO_2 into carbohydrate. Both use H^+ pumps and an H^+ gradient to make ATP.

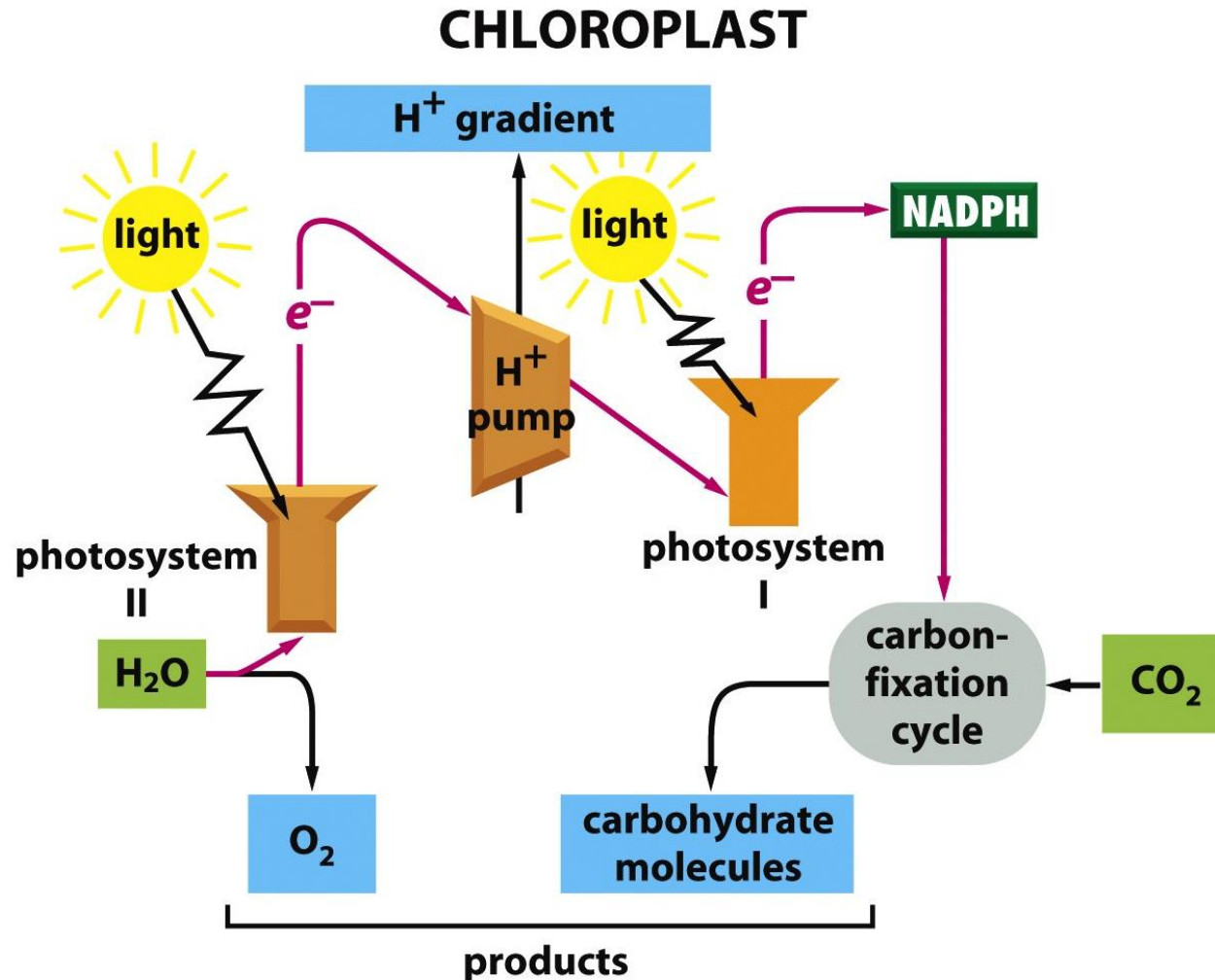
Mitochondrion: food $\rightarrow$ NADH $\rightarrow$ H^+ pumps $\rightarrow$ gradient; O_2 accepts the electrons



KEY POINTS

- Fats and carbohydrates feed the citric acid cycle, which releases CO₂ and generates NADH.
- Electrons from NADH pass through three H⁺ pumps before reducing O₂ to H₂O.
- The H⁺ gradient they build drives ATP synthesis.

Chloroplast: light drives electrons from water to NADPH; CO₂ is fixed into sugar



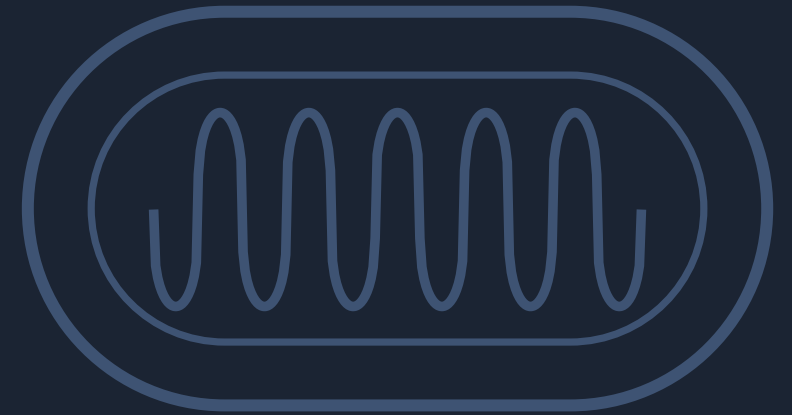
KEY POINTS

- Two photosystems use light energy to move electrons from H₂O (releasing O₂) to NADPH.
- Electron flow between the photosystems pumps H⁺ and builds the gradient for ATP synthesis.
- ATP and NADPH power the carbon-fixation cycle.

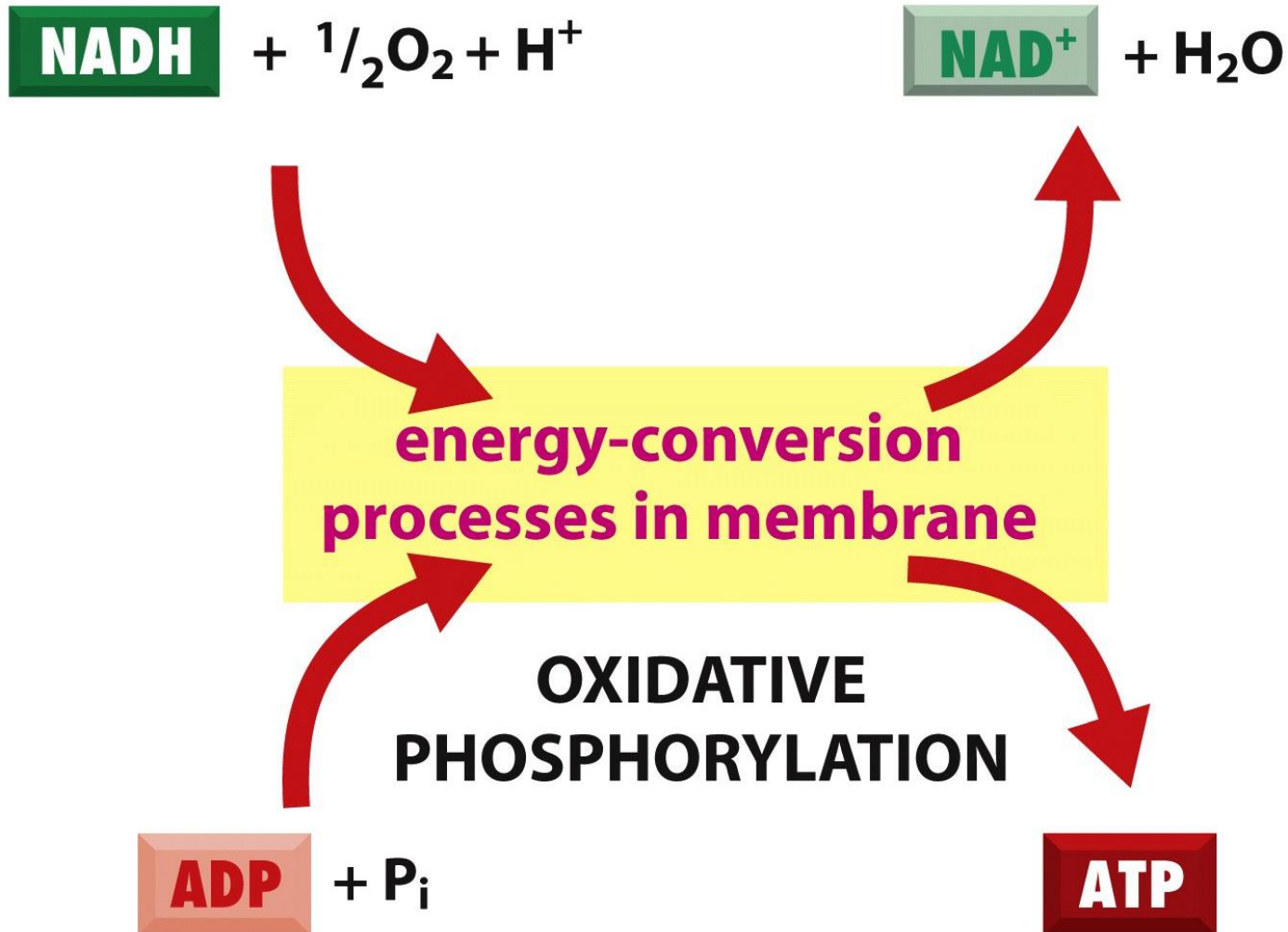
04

How does NADH donate its electrons?

From fuel to high-energy electrons



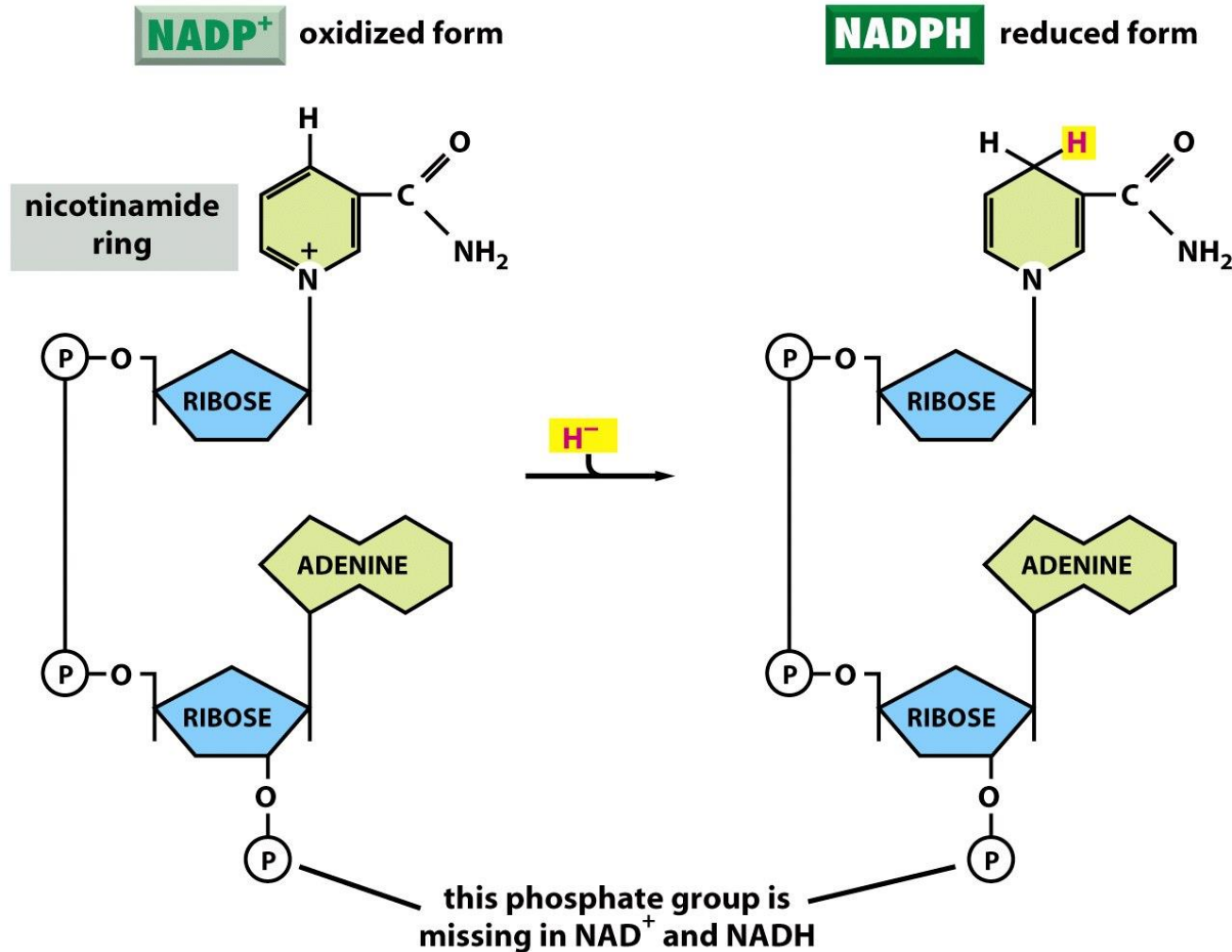
Oxidative phosphorylation couples NADH oxidation to ATP synthesis



KEY POINTS

- **Oxidation** — $\text{NADH} + \frac{1}{2}\text{O}_2 + \text{H}^+ \rightarrow \text{NAD}^+ + \text{H}_2\text{O}$ releases a large amount of free energy.
- **Phosphorylation** — $\text{ADP} + \text{P}_i \rightarrow \text{ATP}$ requires energy.
- **Coupling** — energy-conversion processes in the inner membrane (the H^+ gradient) link the two.

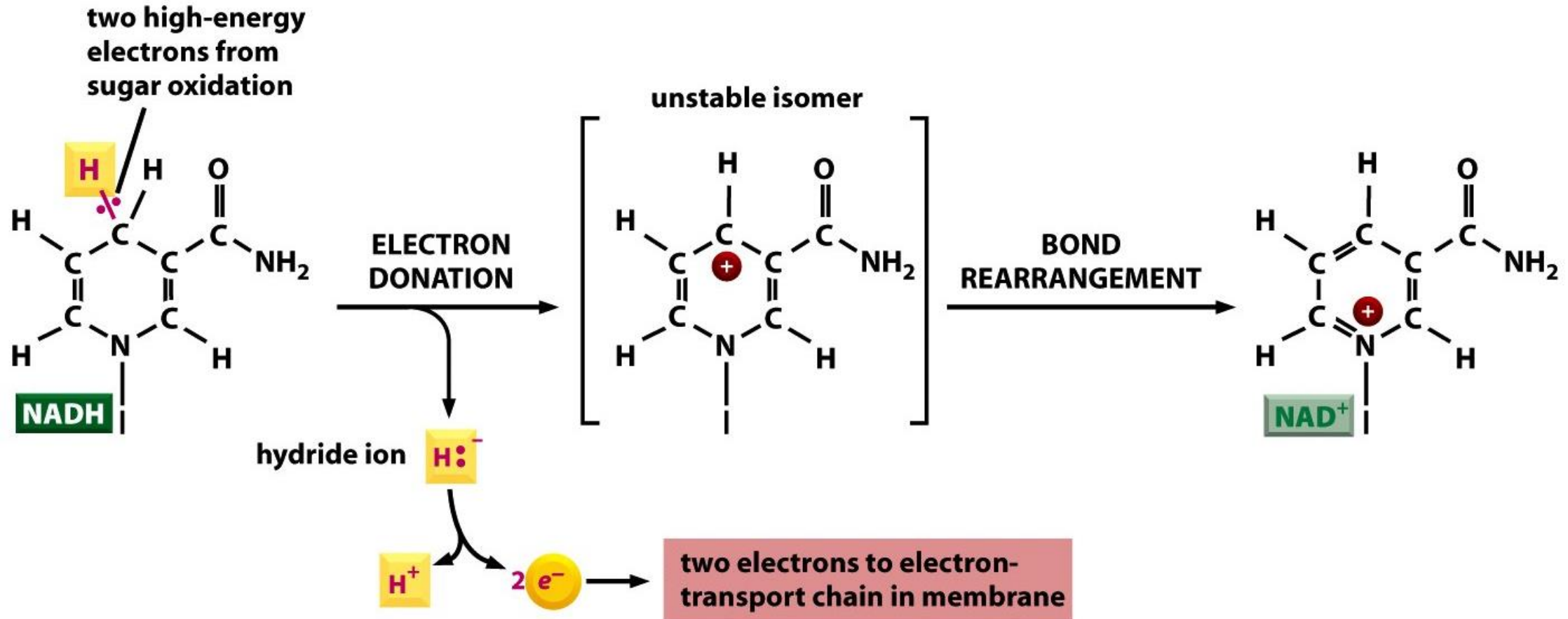
NAD(P)⁺ carries electrons as a hydride ion on its nicotinamide ring



KEY POINTS

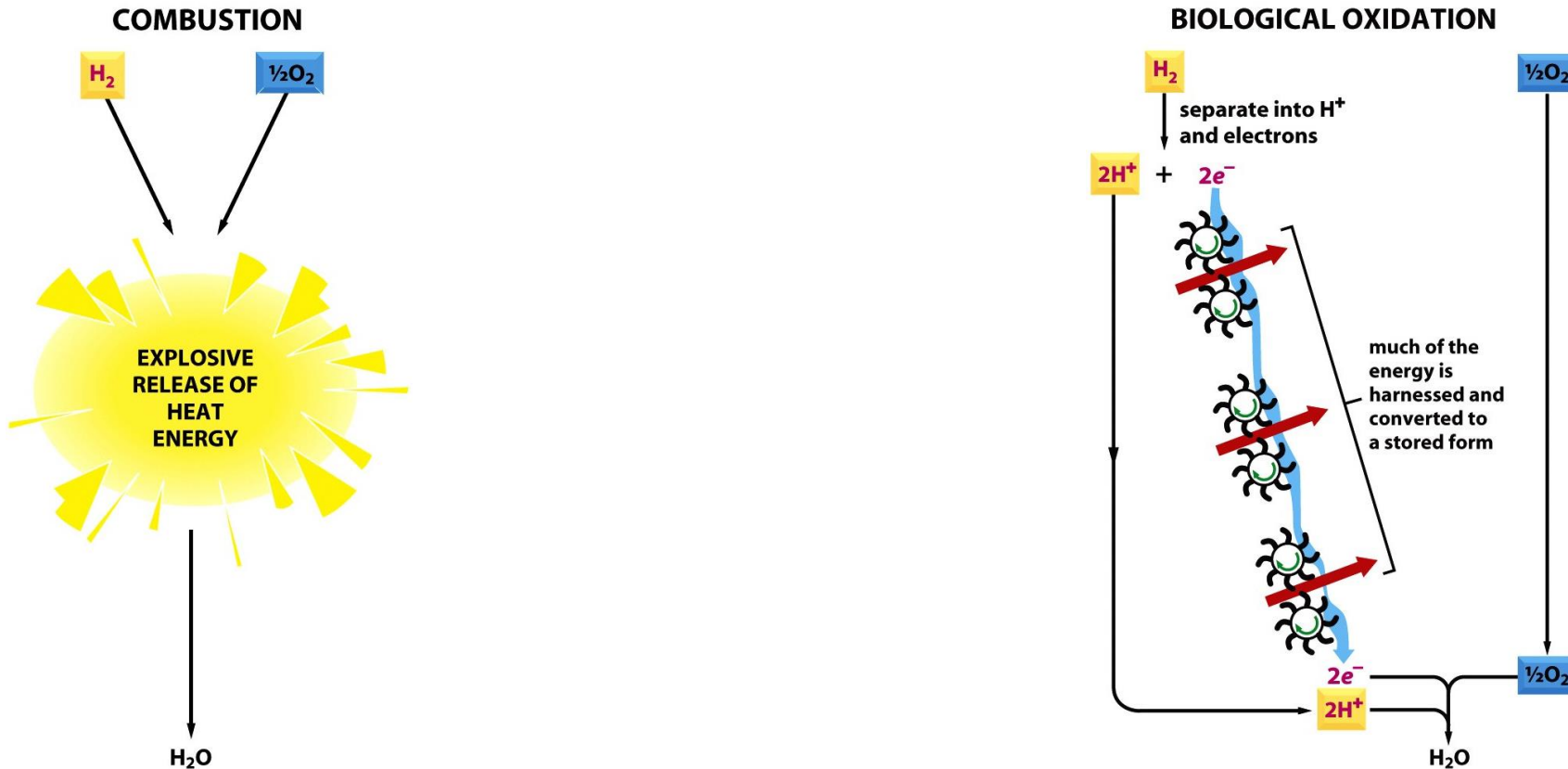
- On reduction the nicotinamide ring accepts a **hydride ion** ($\text{H}^- = \text{H}^+ + 2 \text{e}^-$).
- The figure shows **NADP⁺/NADPH**; NAD⁺/NADH are identical except for the phosphate marked at the bottom.
- NADH feeds catabolic energy into the respiratory chain; NADPH powers biosynthesis.

NADH → NAD⁺: the hydride delivers two high-energy electrons to the chain



The hydride ion of NADH is removed as a proton plus two electrons; the ring rearranges to NAD⁺ and the two electrons are passed to the electron-transport chain in the inner membrane.

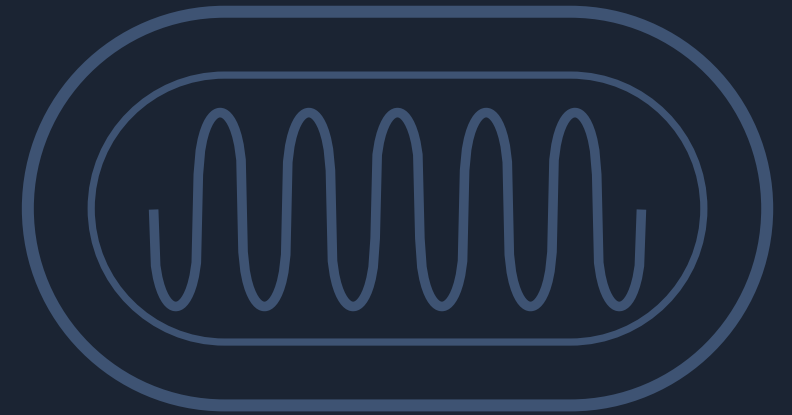
Why not react electrons, H^+ and O_2 directly? Stepwise oxidation captures energy



Combustion (left) releases all the energy at once as heat. Biological oxidation (right) splits H_2 into H^+ and electrons, which pass through a series of carriers so that much of the energy is harnessed in a stored form.

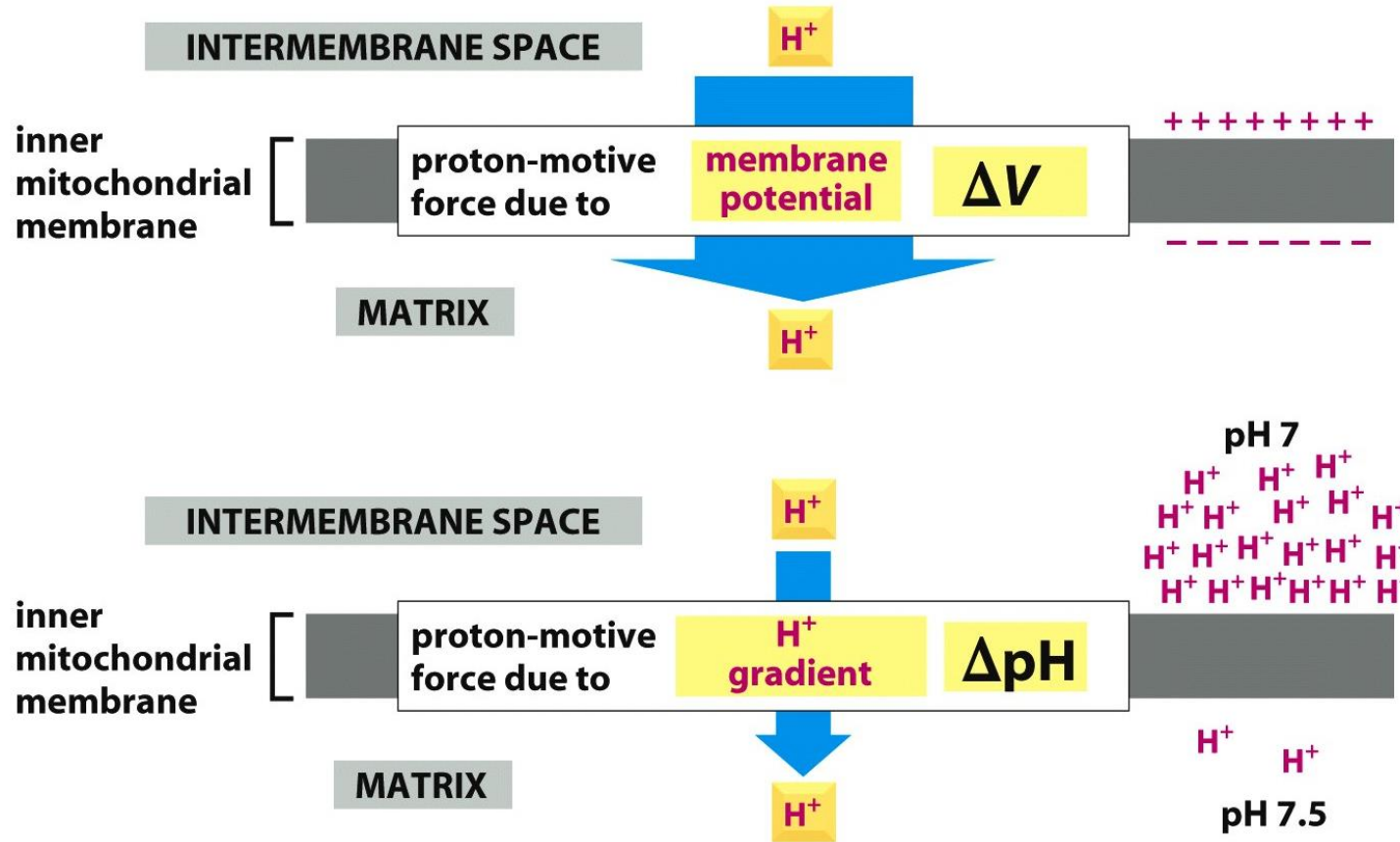
05

What are the two forces of the proton gradient — and what do they do to protons?



Membrane potential and pH gradient

The proton-motive force = membrane potential (ΔV) + pH gradient (ΔpH)



PROTON-MOTIVE FORCE (Δp)

$\approx 180\text{--}190\text{ mV}$

ΔV 160–170 mV

ΔpH 20–30 mV

- **Membrane potential (ΔV)** — matrix negative relative to the intermembrane space
- **pH gradient (ΔpH)** — matrix ~ 0.5 pH unit more alkaline; 1 pH unit $\approx 61\text{ mV}$ at 37°C
- **Both forces push H^+ into the matrix** — through ATP synthase or H^+ -coupled carriers

Break

KEYWORDS SO FAR

Chemiosmosis

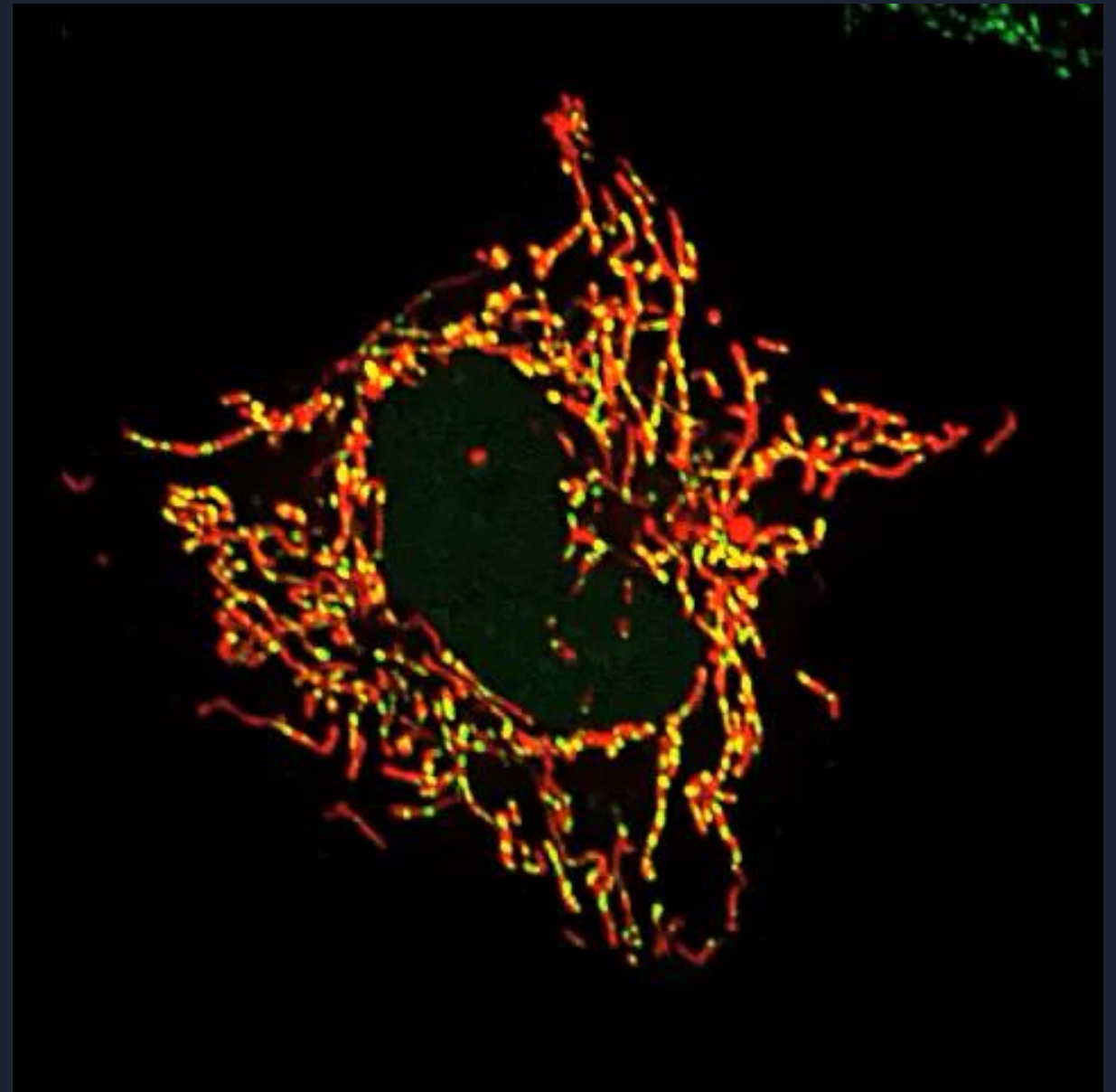
ATP synthesis driven by an H^+ gradient that electron transport creates

Electrochemical proton gradient

$\Delta V + \Delta pH$ across the inner membrane (≈ 180 mV); pushes H^+ into the matrix

Electron-transport chain

Inner-membrane complexes that pass electrons from NADH to O_2 and pump H^+

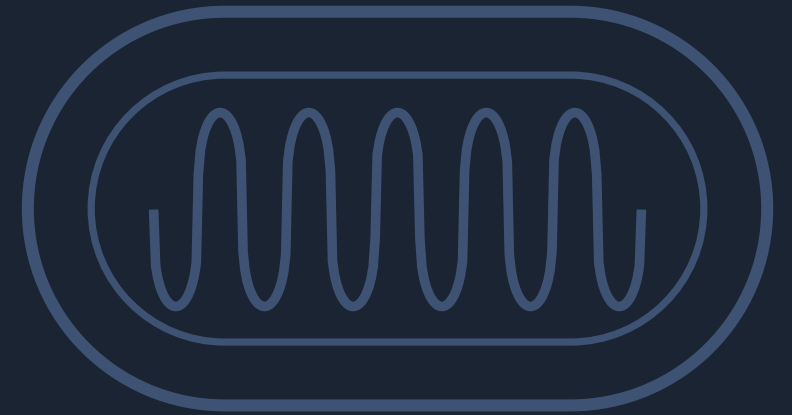


Video: mitochondrial nucleoid dynamics (Phys.org / Medical Xpress)

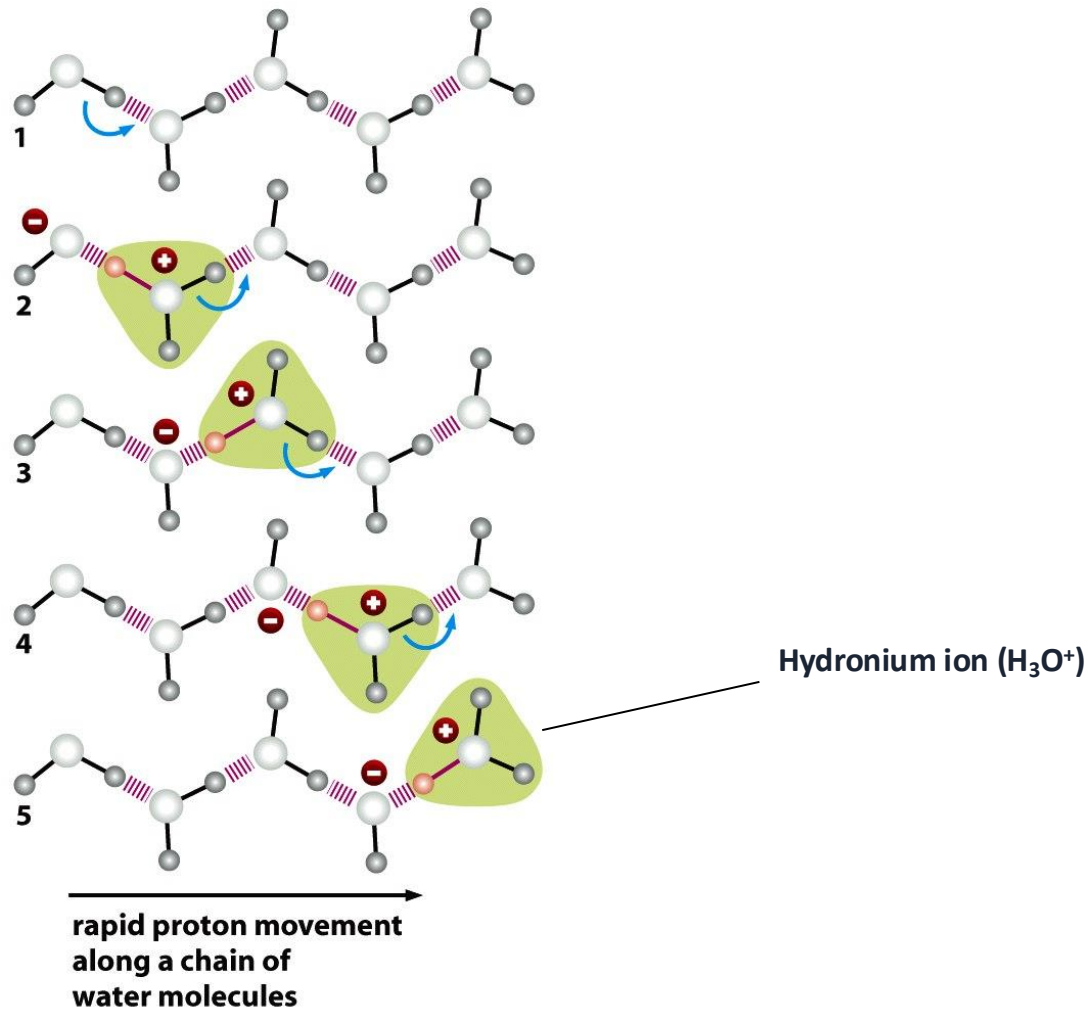
06

Why use protons for the electron-transport chain?

Rapid H^+ transfer through water and proteins



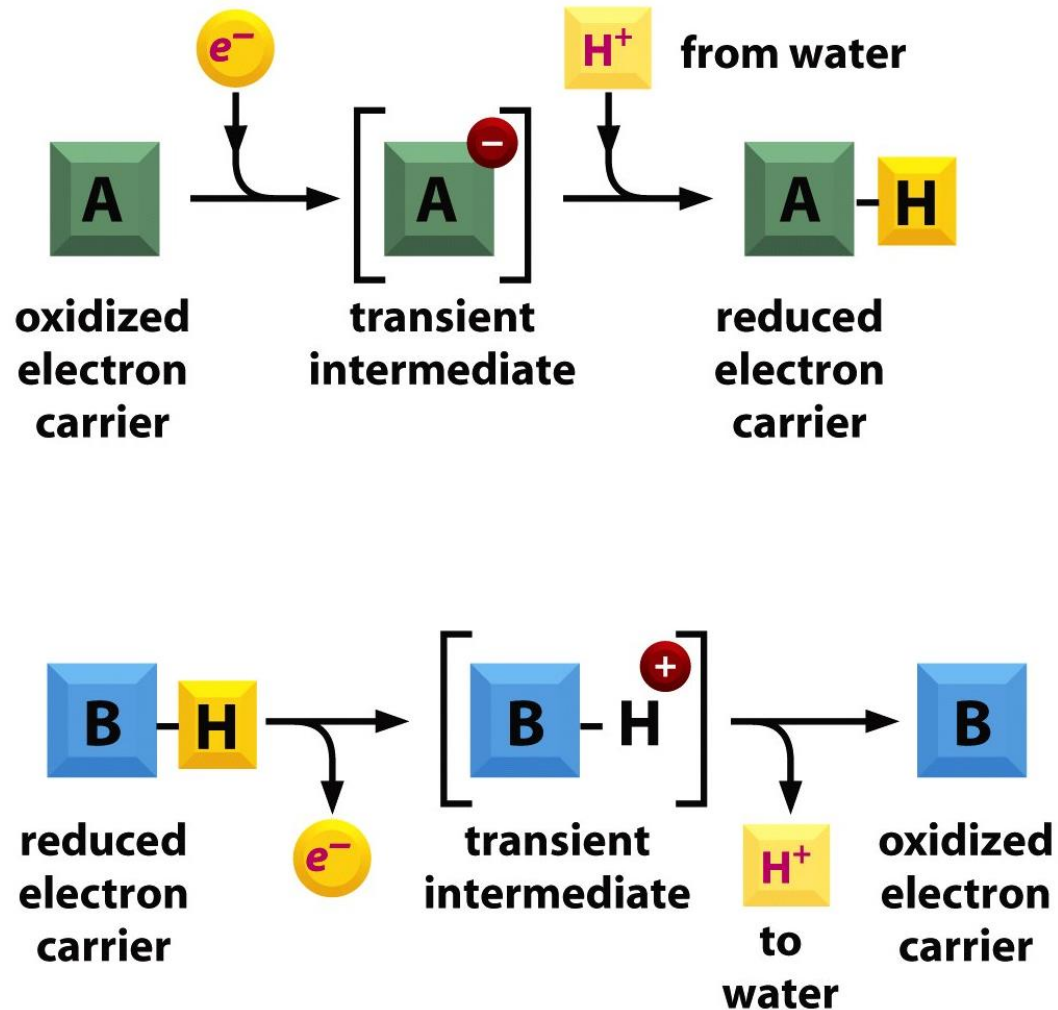
Protons hop rapidly along hydrogen-bond chains in water and proteins



KEY POINTS

- A proton in water exists as **hydronium** (H_3O^+), not as a free ion.
- It jumps to a neighbouring water molecule, which passes another proton on — a relay (**Grotthuss mechanism**), much faster than diffusion of an ion.
- Chains of water and polar side chains inside membrane proteins let pumps move H^+ quickly.

Electron carriers pick up and release whole protons from water

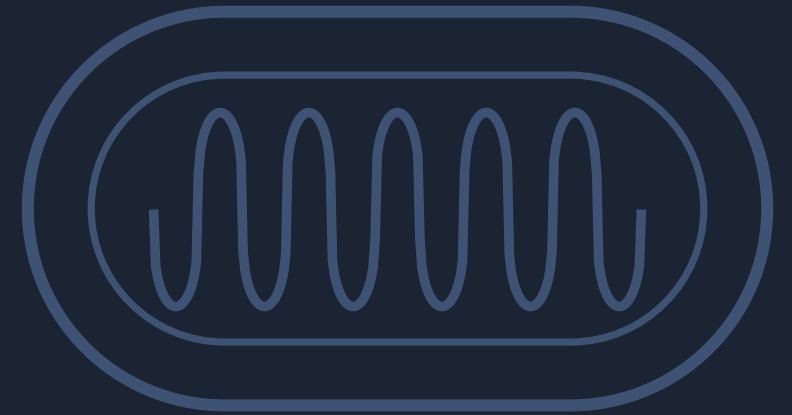


KEY POINTS

- **(A)** Adding an electron to carrier A creates a charged intermediate that grabs an H^+ from water $\rightarrow$ reduced AH.
- **(B)** Removing the electron from BH makes the H^+ leave to water.
- If uptake and release happen on opposite membrane faces, electron transfer moves H^+ across the membrane.

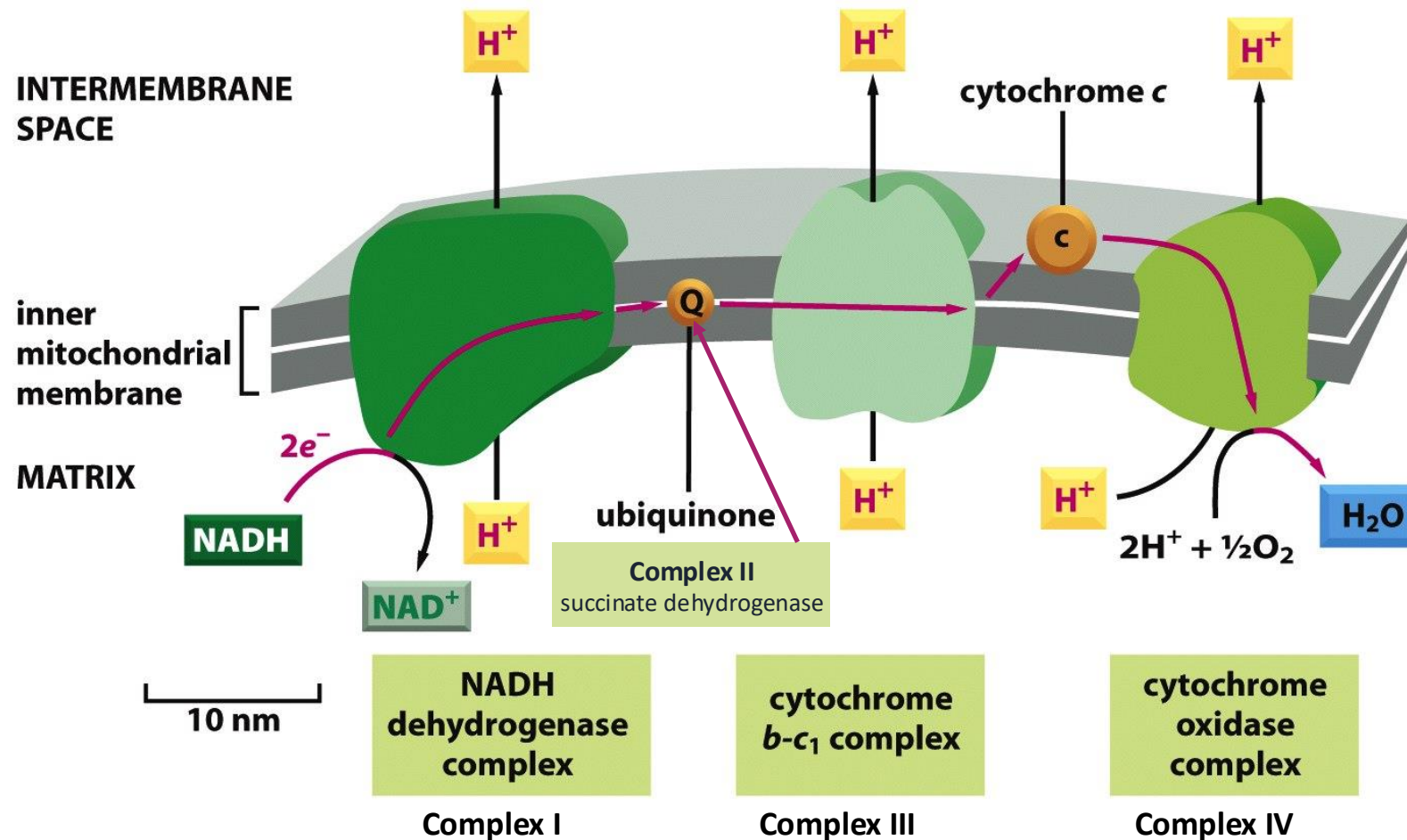
07

How are electrons moved along the respiratory chain?



Complexes I–IV, their electron carriers and proton pumping

Four complexes carry electrons to O₂; three of them (I, III, IV) move H⁺



KEY POINTS

- **Complex II** (succinate dehydrogenase) passes electrons from succinate, via FADH₂, to ubiquinone — but pumps no H⁺.
- **Mobile carriers** — ubiquinone (in the membrane) and cytochrome c (in the intermembrane space).
- **H⁺ moved per NADH (2 e⁻):** I = 4, III = 4, IV = 2 → ≈ 10 H⁺. Via complex II (FADH₂): ≈ 6 H⁺.

Electrons travel on metal centres fixed in proteins and on two mobile carriers

FIXED IN PROTEINS



Heme (cytochromes)

Fe in a porphyrin ring cycles $\text{Fe}^{3+} \rightleftharpoons \text{Fe}^{2+} \cdot 1 \text{ e}^-$ · complexes III and IV, and cytochrome c



Iron–sulfur clusters

$2\text{Fe}-2\text{S}$ or $4\text{Fe}-4\text{S}$ held by cysteines · 1 e^- · complexes I, II and III



Copper centres

Cu_A and Cu_B in cytochrome oxidase · $\text{Cu}^{2+} \rightleftharpoons \text{Cu}^+ \cdot 1 \text{ e}^-$

MOBILE CARRIERS



Ubiquinone (coenzyme Q)

Lipid-soluble, diffuses within the inner membrane · carries $2 \text{ e}^- + 2 \text{ H}^+$ · from complexes I and II to complex III

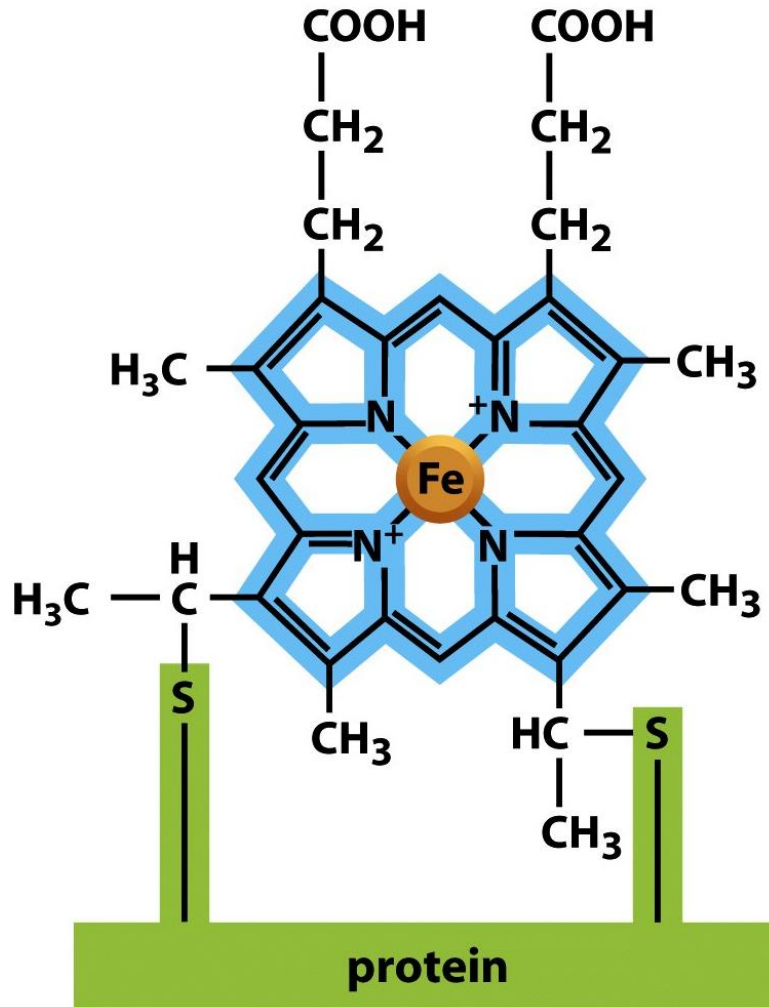


Cytochrome c

Small soluble heme protein in the intermembrane space · carries 1 e^- · from complex III to complex IV

Flavins (FMN in complex I, FAD in complex II) accept the electrons at the entry points of the chain.

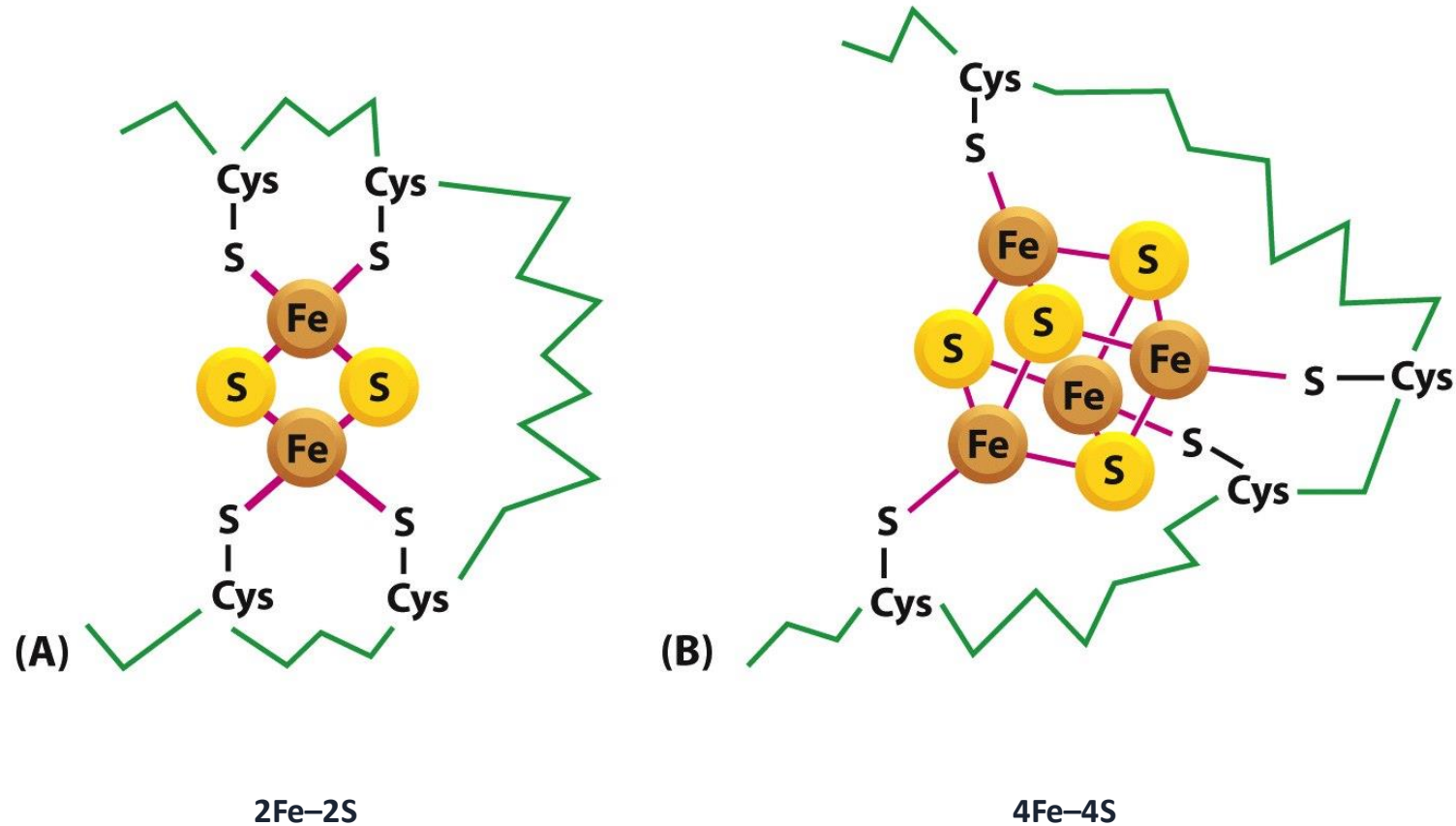
Heme: an iron atom in a porphyrin ring — covalently bound in cytochrome c



KEY POINTS

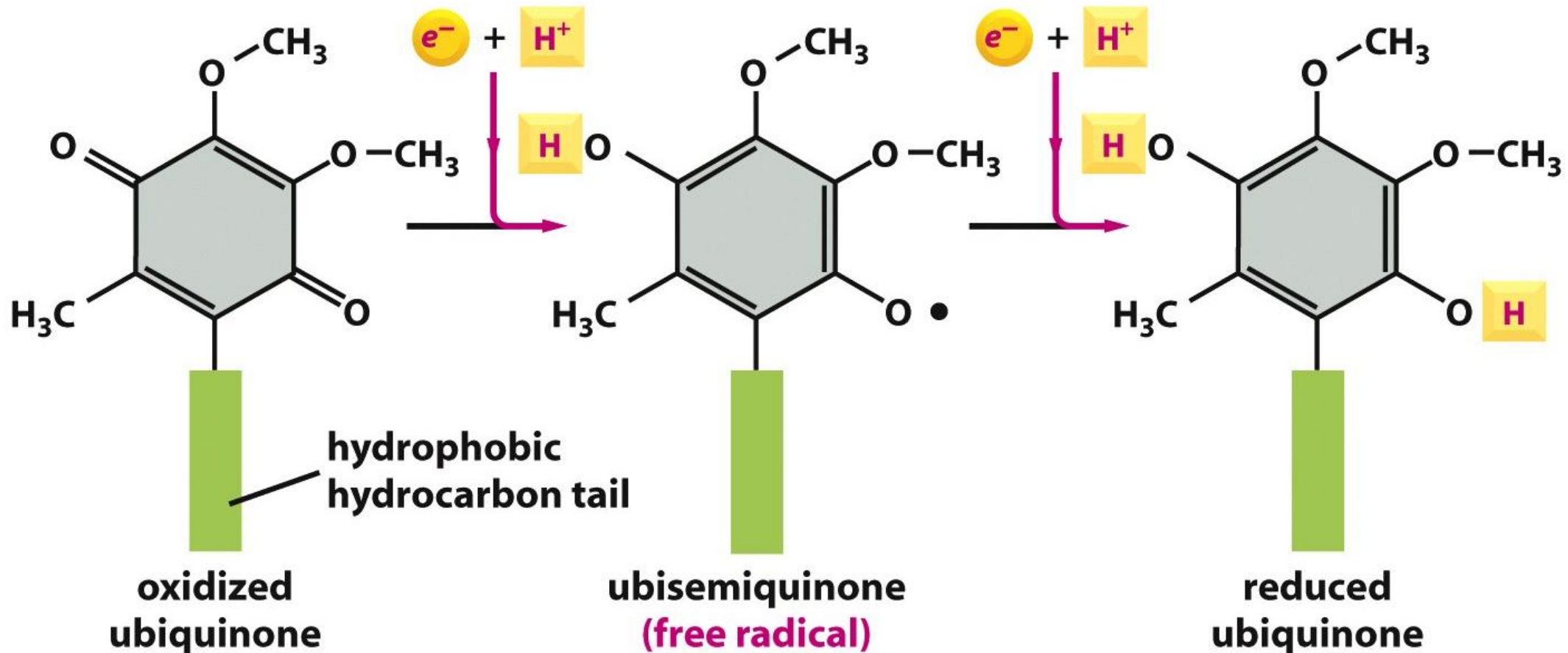
- The Fe atom cycles between Fe³⁺ and Fe²⁺, carrying one electron at a time.
- The protein environment tunes each cytochrome's redox potential (a-, b- and c-type hemes).
- In cytochrome c the heme is attached to the protein through two cysteine thioether bonds.

Iron–sulfur clusters: iron held by cysteine sulfurs, carrying one electron



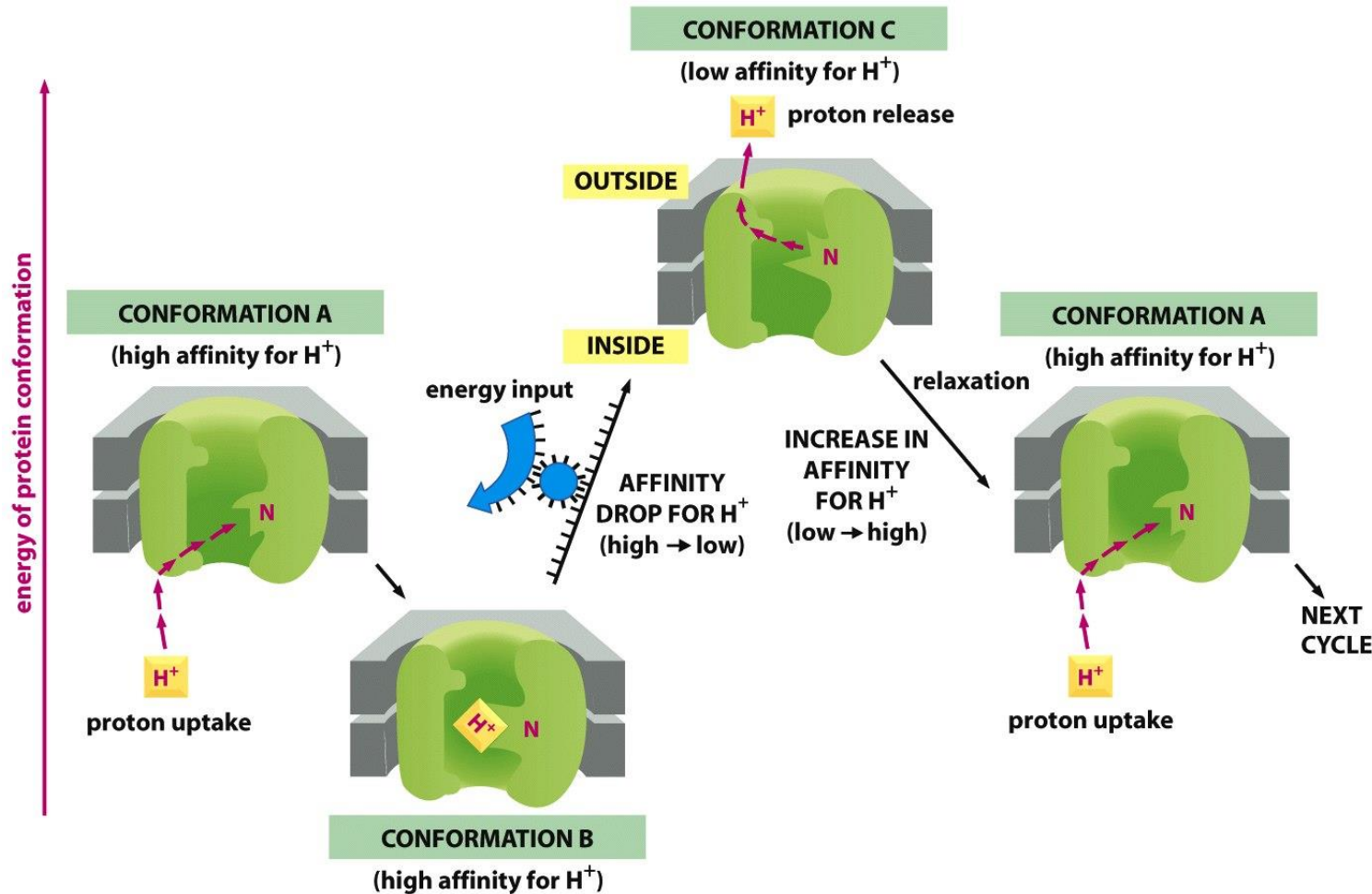
(A) 2Fe–2S and (B) 4Fe–4S clusters. Iron (Fe) is coordinated by inorganic sulfide (S) and by the sulfur of cysteine side chains in the protein. Whatever its size, a cluster transfers one electron at a time.

Ubiquinone carries two electrons and two H⁺ — one electron at a time



Oxidised ubiquinone (Q) takes up one electron + H⁺ to form the semiquinone radical, then a second to form ubiquinol (QH₂). The hydrophobic tail keeps it diffusing within the inner membrane; the semiquinone is a major source of superoxide.

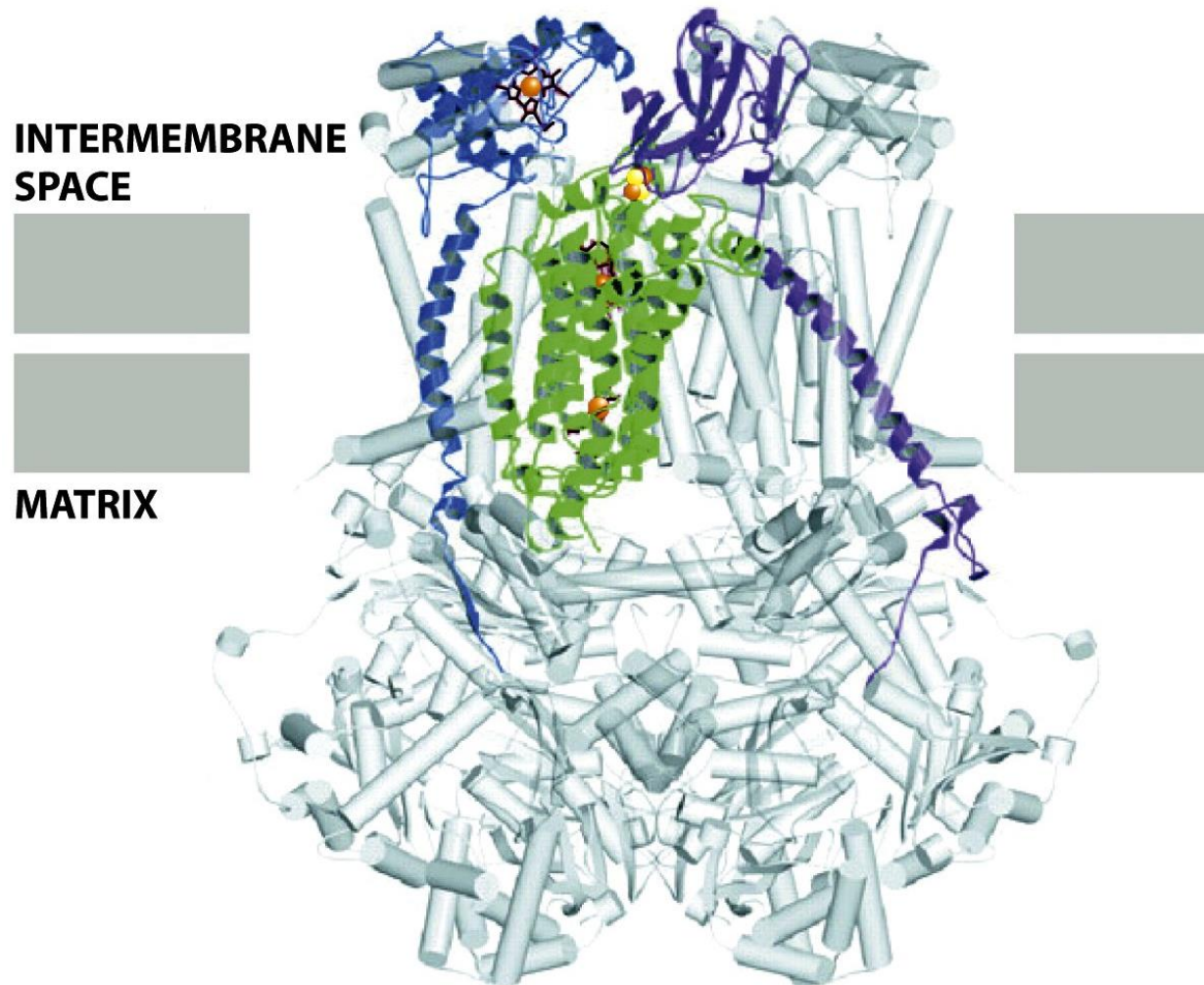
Proton pumping: conformational changes switch H^+ affinity and membrane side



KEY POINTS

- **A → B** — the site faces the matrix and binds H^+ with high affinity.
- **B → C** — energy from electron transfer drives a conformational change: the site now faces outside and its affinity drops.
- **C → A** — H^+ is released to the intermembrane space; the protein relaxes, ready for the next cycle.
- General model for complexes I and IV.

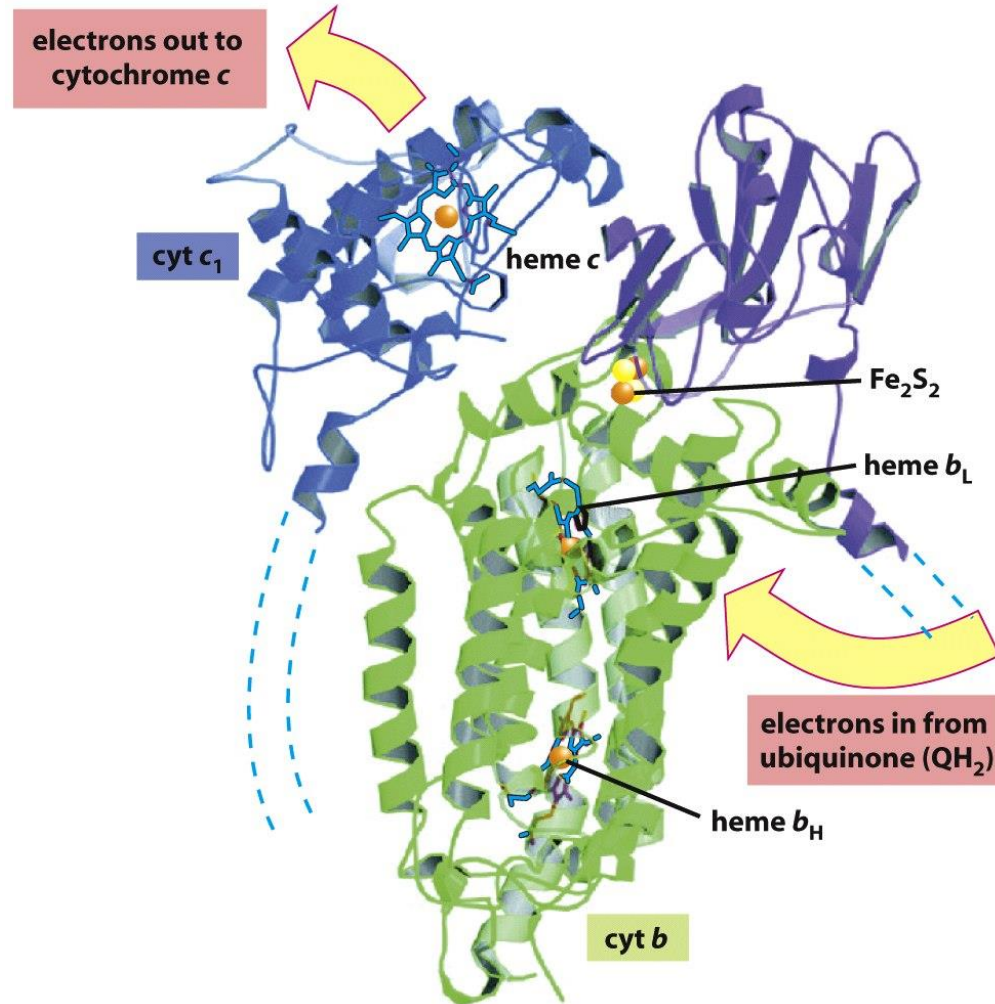
Cytochrome b-c₁ complex (Complex III) is a dimer spanning the inner membrane



KEY POINTS

- Each monomer contains cytochrome b (two hemes), cytochrome c₁ and an iron–sulfur (Rieske) protein.
- It takes electrons from ubiquinol (QH₂) in the membrane and passes them to cytochrome c in the intermembrane space.

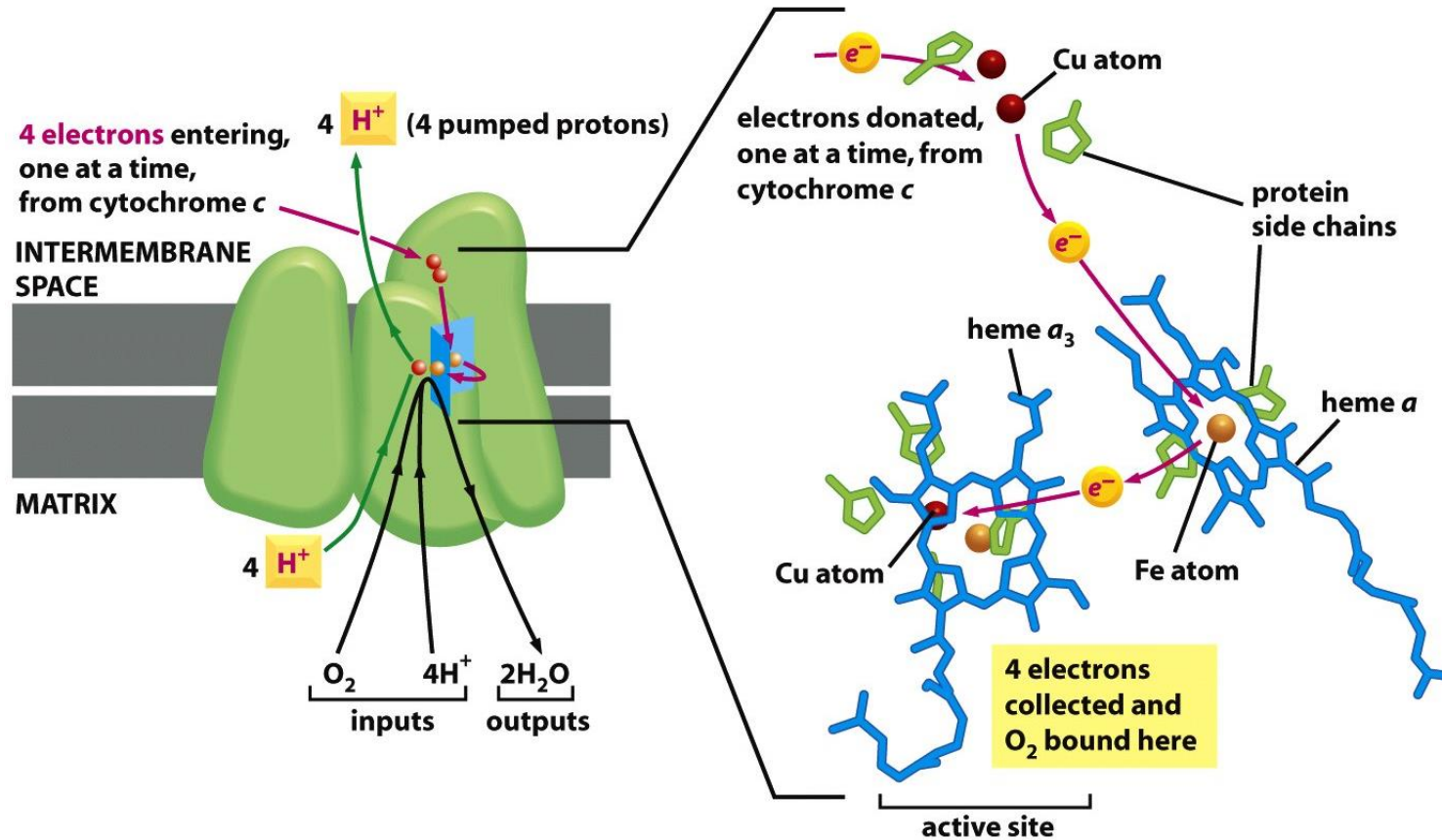
Complex III moves H^+ via the Q cycle rather than a conformational pump



THE Q CYCLE

- Ubiquinol (QH_2) releases its 2 H^+ to the intermembrane space and gives up 2 e^- .
- One e^- goes via the Fe-S centre and cyt c_1 to cytochrome c .
- The other recycles through hemes b_L and b_H to reduce a Q on the matrix side, which takes up matrix H^+ .
- Net per 2 e^- to cyt c :** 4 H^+ released outside, 2 H^+ taken from the matrix.

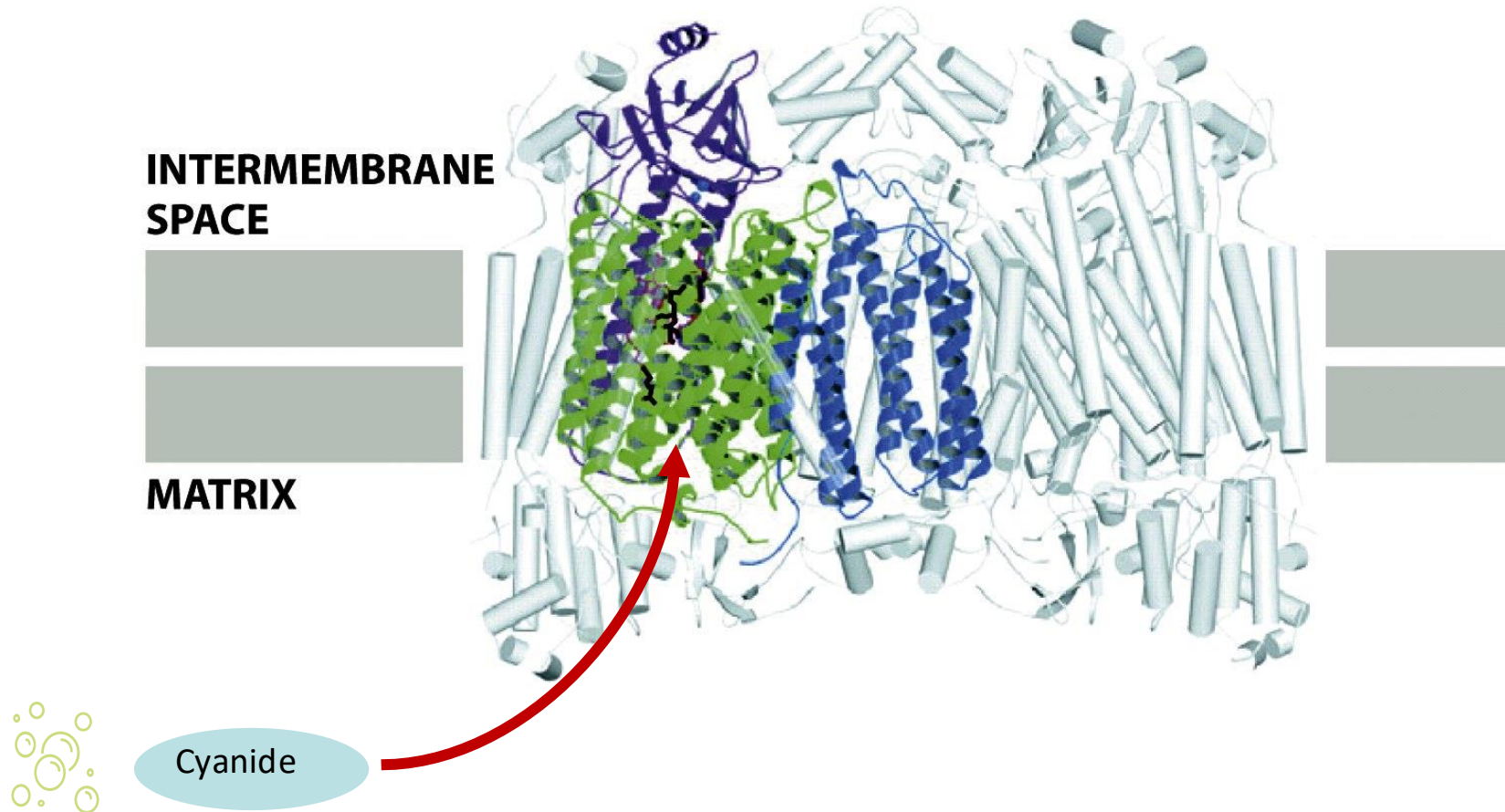
Cytochrome oxidase (Complex IV) collects four electrons to reduce O_2 to water



KEY POINTS

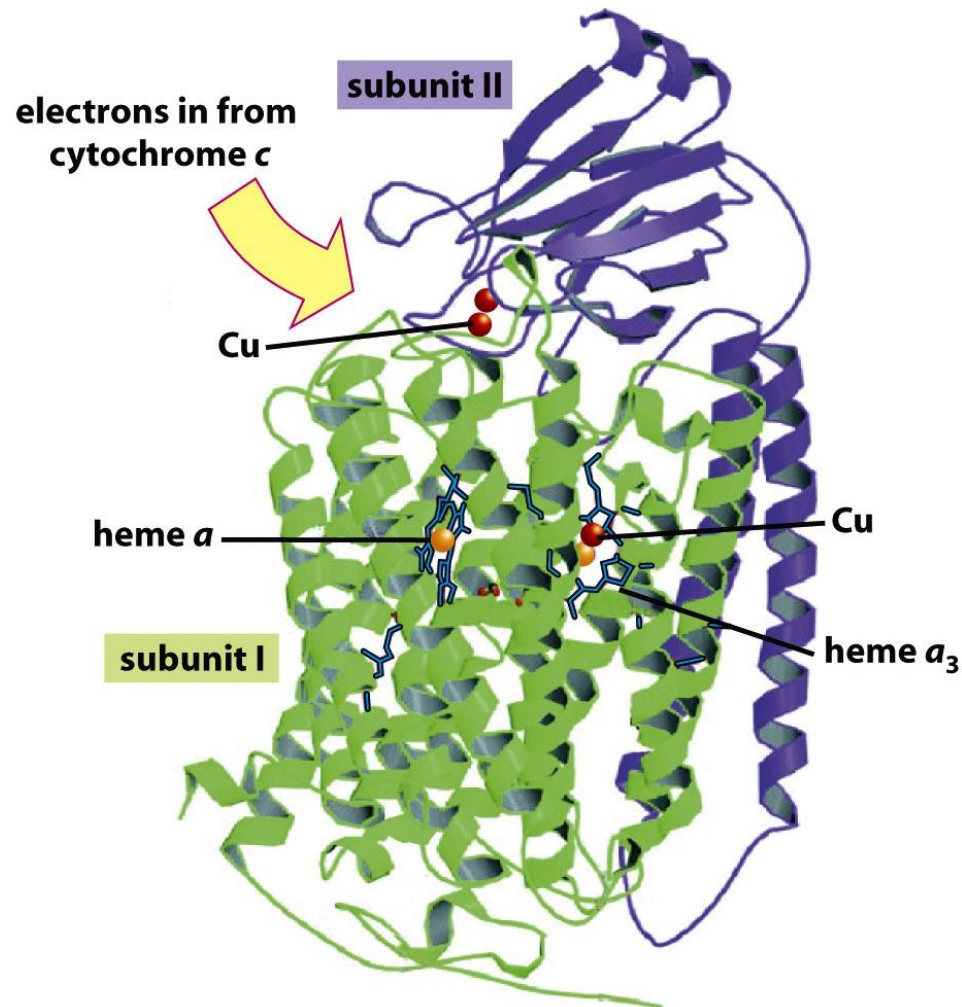
- $4 e^-$ arrive one at a time from cytochrome c, via a copper centre and heme a.
- They are collected at the heme a_3 -Cu active site, where O_2 stays bound until fully reduced to $2 H_2O$.
- **Per O_2 :** 4 "chemical" H^+ taken from the matrix + 4 H^+ pumped across the membrane.

Cytochrome oxidase: a large complex around a compact core — cyanide's target



Structure of the multi-subunit complex in the inner membrane; the functional core (coloured) carries the metal centres. Cyanide binds the heme a₃-Cu_B site, blocking O₂ reduction and halting the entire chain.

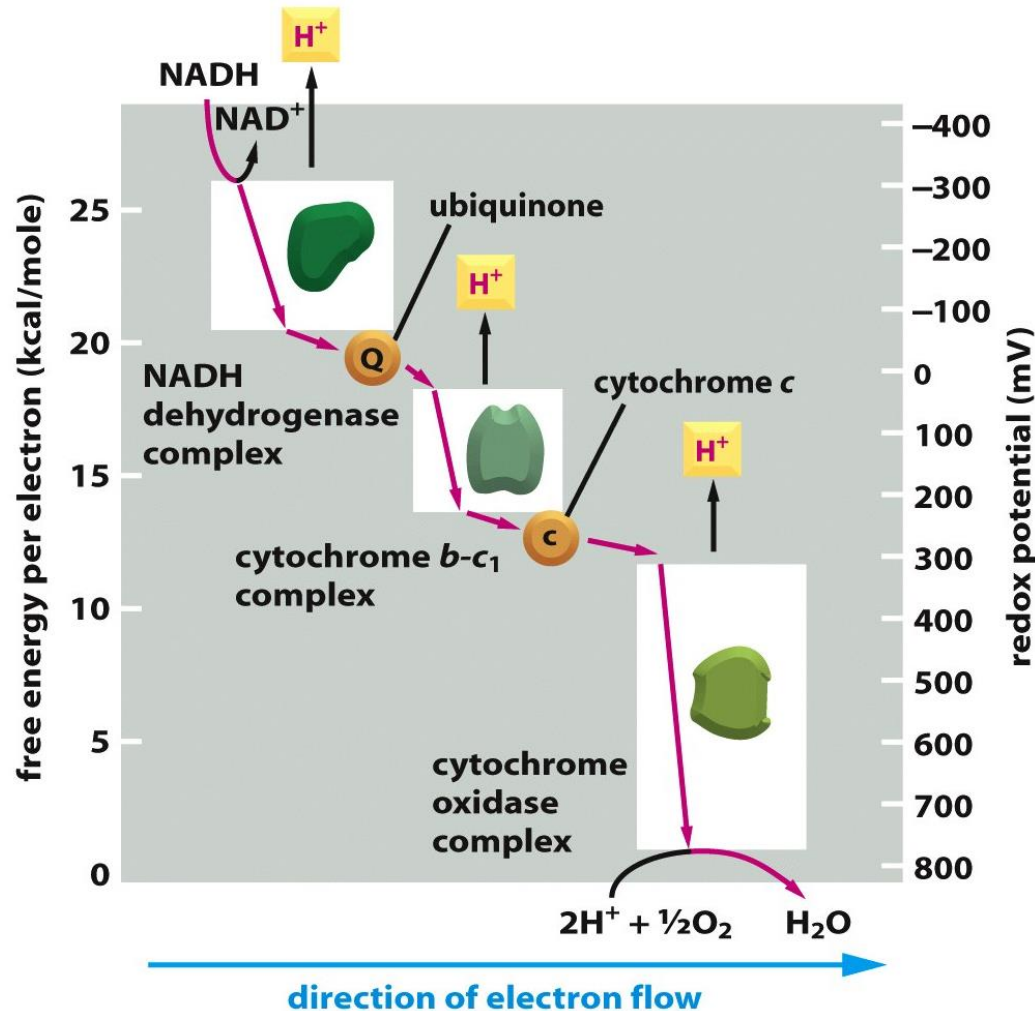
Catalytic core: electrons pass $\text{Cu}_A \rightarrow \text{heme } a \rightarrow \text{heme } a_3\text{--Cu}_B \rightarrow \text{O}_2$



KEY POINTS

- **Subunit II** receives electrons from cytochrome c at its copper centre (Cu_A).
- **Subunit I** holds heme a and the binuclear heme $a_3\text{--Cu}_B$ centre, where O_2 is reduced.

Electrons move “downhill” in redox potential, releasing free energy in steps

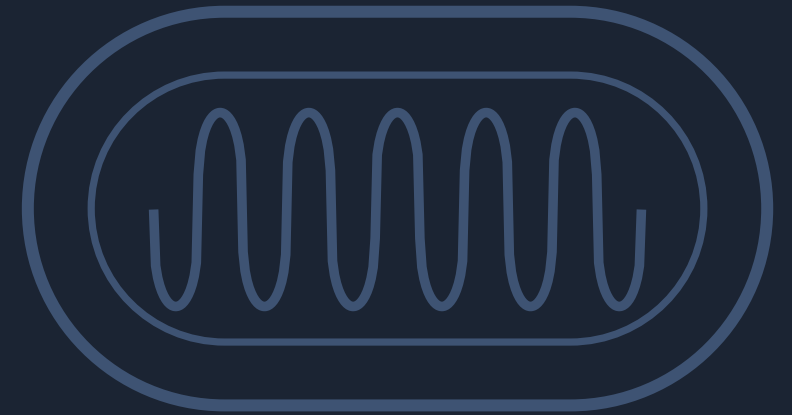


KEY POINTS

- Redox potential rises from NADH (≈ -320 mV) to O_2 ($\approx +820$ mV): each carrier has a stronger tendency to accept electrons than the one before.
- The free energy released at the three large steps is used to pump H^+ (complexes I, III and IV).

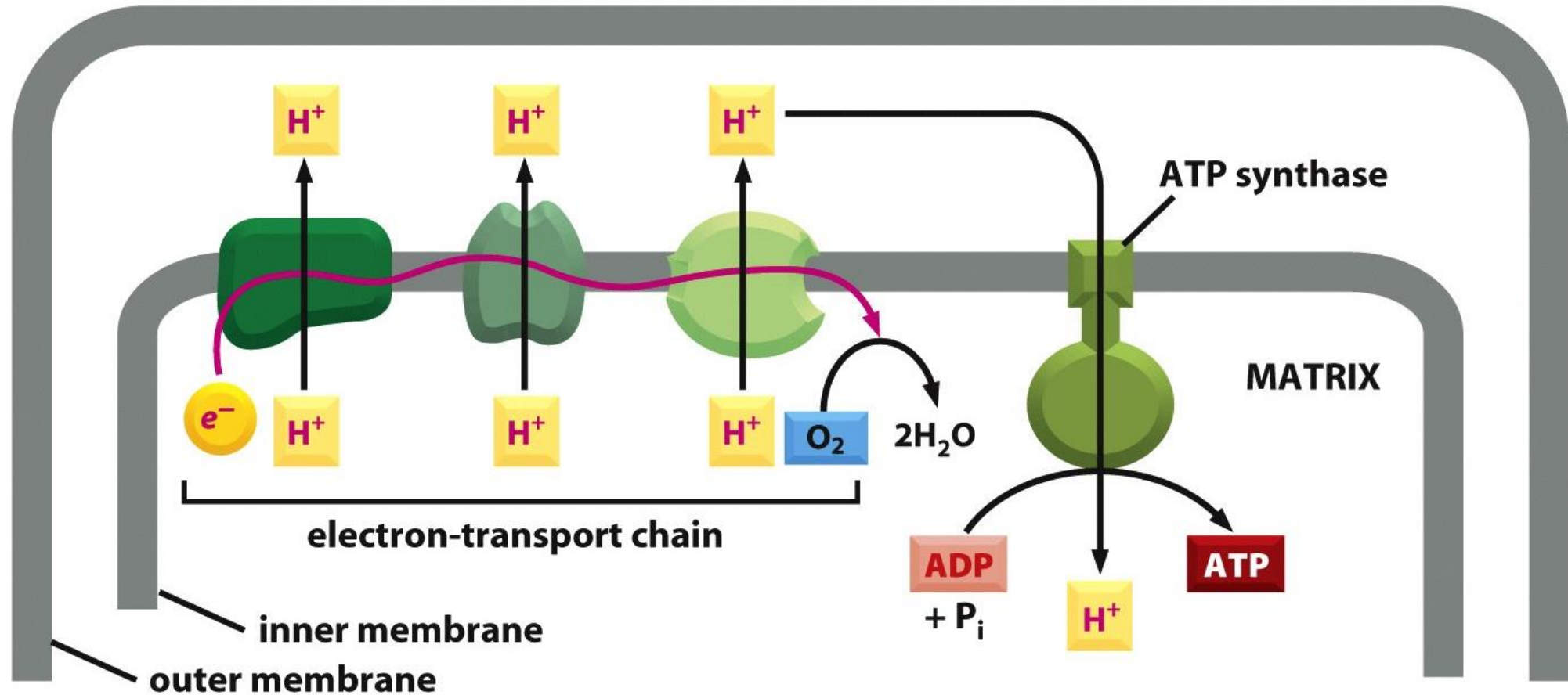
08

How is the proton gradient turned into ATP?



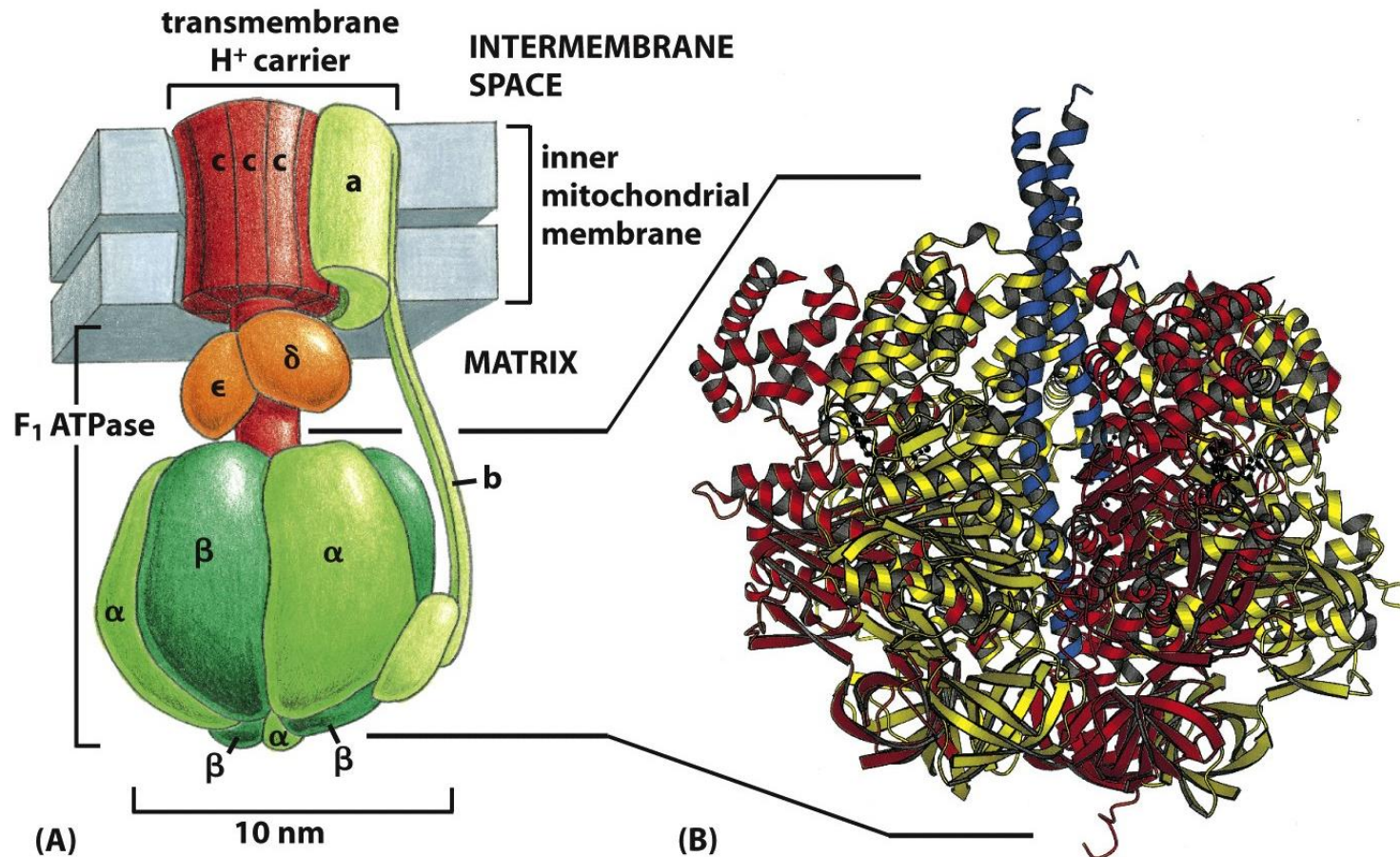
ATP synthase, reversibility and gradient-driven transport

Recap: the H^+ gradient built by the chain drives ATP synthase



Electron transport pumps H^+ into the intermembrane space; H^+ flow back into the matrix through ATP synthase, which makes ATP from ADP + P_i .

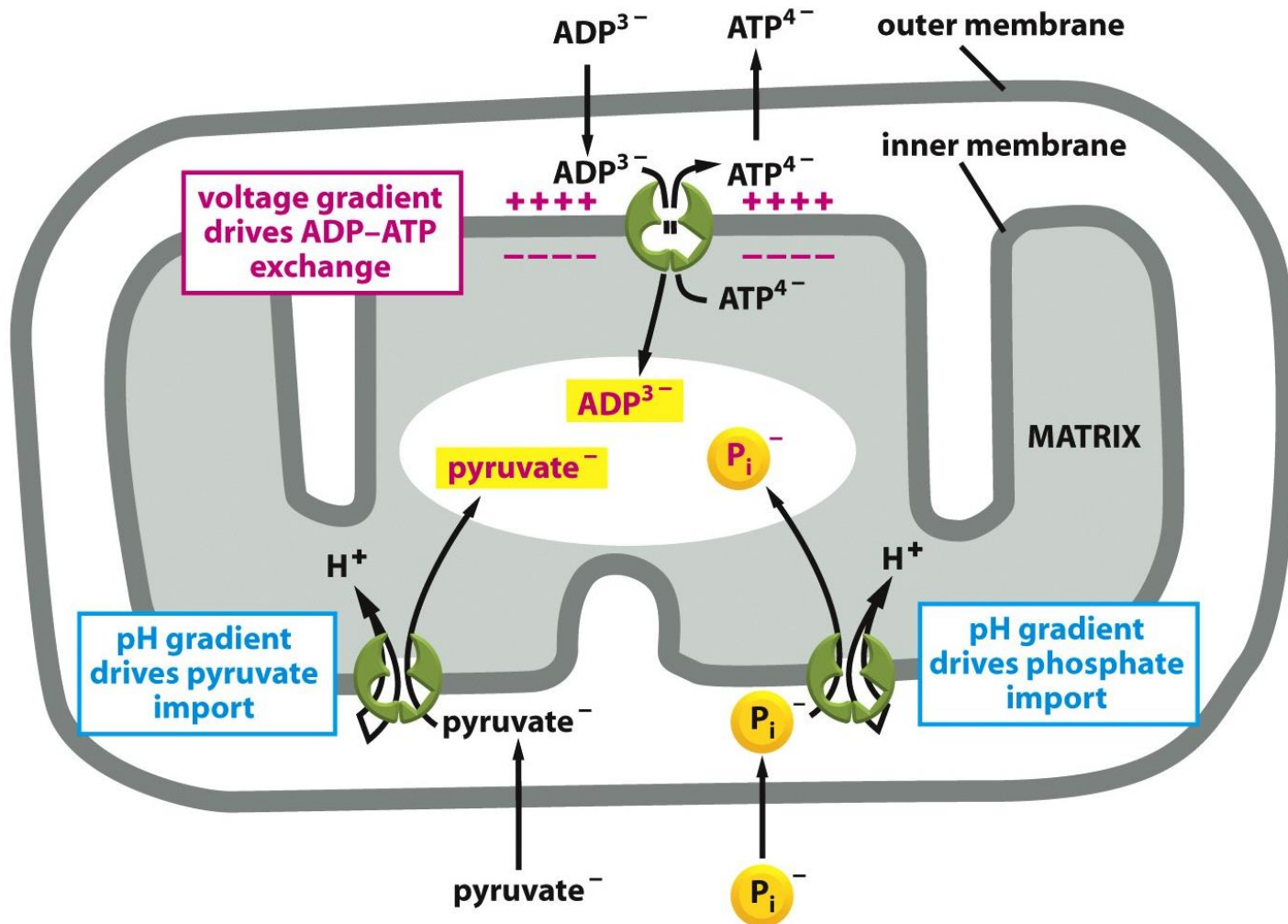
ATP synthase is a rotary motor: ~3 ATP per turn, up to ~100 ATP per second



KEY POINTS

- **F₀ (membrane)** — H⁺ flowing through subunit a turns the ring of c subunits and the central stalk.
- **F₁ head (matrix, $\alpha_3\beta_3$)** — the rotating stalk drives conformational changes in the β subunits that make ATP.
- **Peripheral stalk (b)** — a stator that holds the head still against rotation.

The same gradient drives metabolite transport across the inner membrane



KEY POINTS

- **Voltage gradient** drives ADP^{3-} in / ATP^{4-} out exchange by the ADP/ATP carrier (one net negative charge exported).
- **pH gradient** drives import of pyruvate and phosphate together with H^+ .
- The outer membrane is freely permeable to these small molecules (porins).

ΔG depends on ΔG° and on concentrations: cells keep ATP far from equilibrium

For the reaction



the following equation applies:

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{ADP}] [\text{P}_i]}{[\text{ATP}]}$$

where ΔG and ΔG° are in kilocalories per mole, R is the gas constant (2×10^{-3} kcal/mole K), T is the absolute temperature (K), and all the concentrations are in moles per liter.

When the concentrations of all reactants are at 1 M, $\Delta G = \Delta G^\circ$ (since $RT \ln 1 = 0$). ΔG° is thus a **constant** defined as the **standard free-energy change** for the reaction.

At equilibrium the reaction has no net effect on the disorder of the universe, so $\Delta G = 0$. Therefore, at equilibrium,

$$-RT \ln \frac{[\text{ADP}] [\text{P}_i]}{[\text{ATP}]} = \Delta G^\circ$$

But the concentrations of reactants at equilibrium must satisfy the equilibrium equation:

$$\frac{[\text{ADP}] [\text{P}_i]}{[\text{ATP}]} = K$$

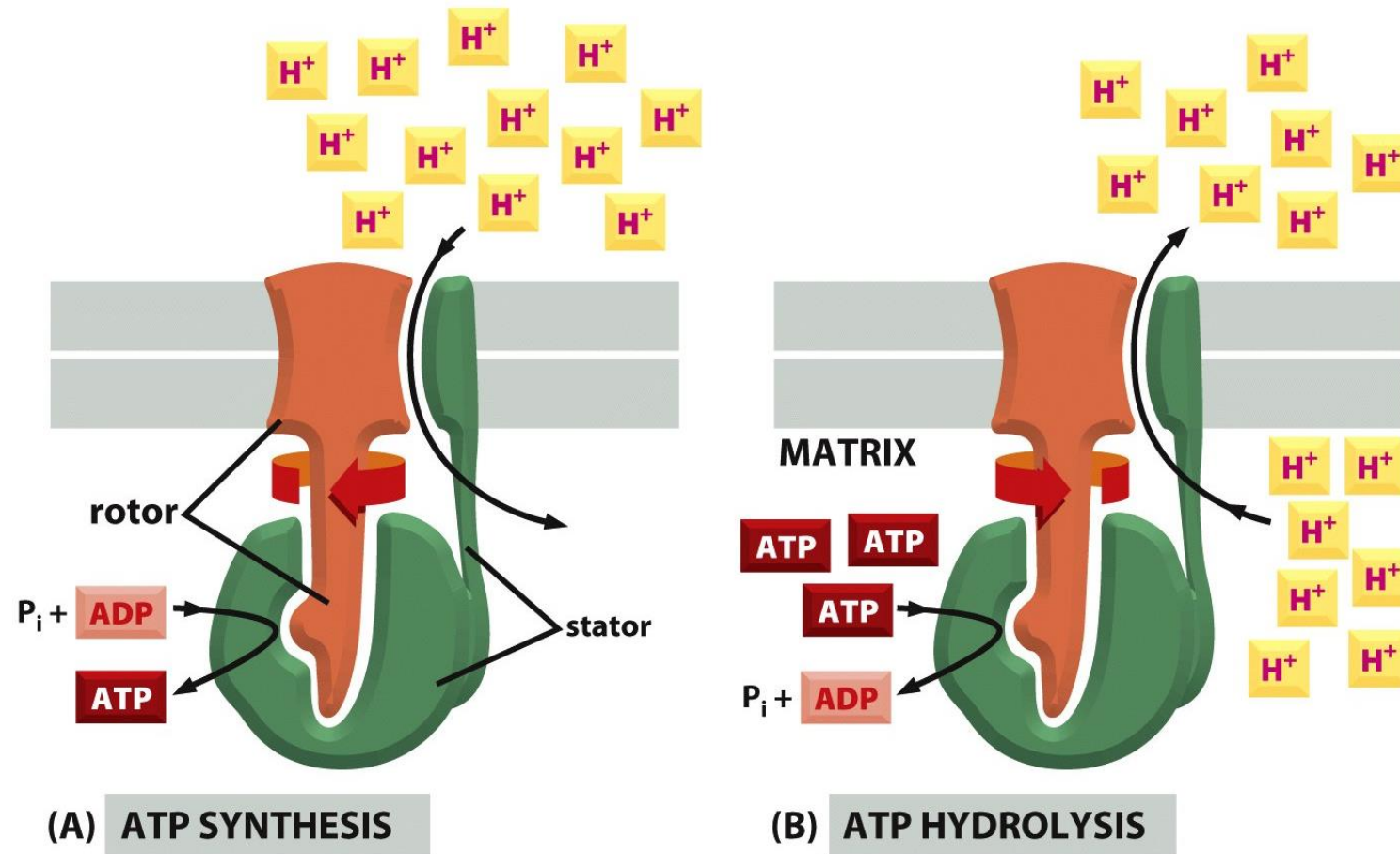
Therefore, at equilibrium,

$$\Delta G^\circ = -RT \ln K$$

We thus see that whereas ΔG° indicates the equilibrium point for a reaction, ΔG reveals *how far the reaction is from equilibrium*. ΔG is a measure of the “driving force” for any chemical reaction, just as the proton-motive force is the driving force for the translocation of protons.

ΔG° sets the equilibrium point; ΔG says how far a reaction is from it. Cells keep $[\text{ATP}]/[\text{ADP}][\text{P}_i]$ far above equilibrium, so ATP hydrolysis releases about -11 to -13 kcal/mol in vivo versus -7.3 kcal/mol under standard conditions.

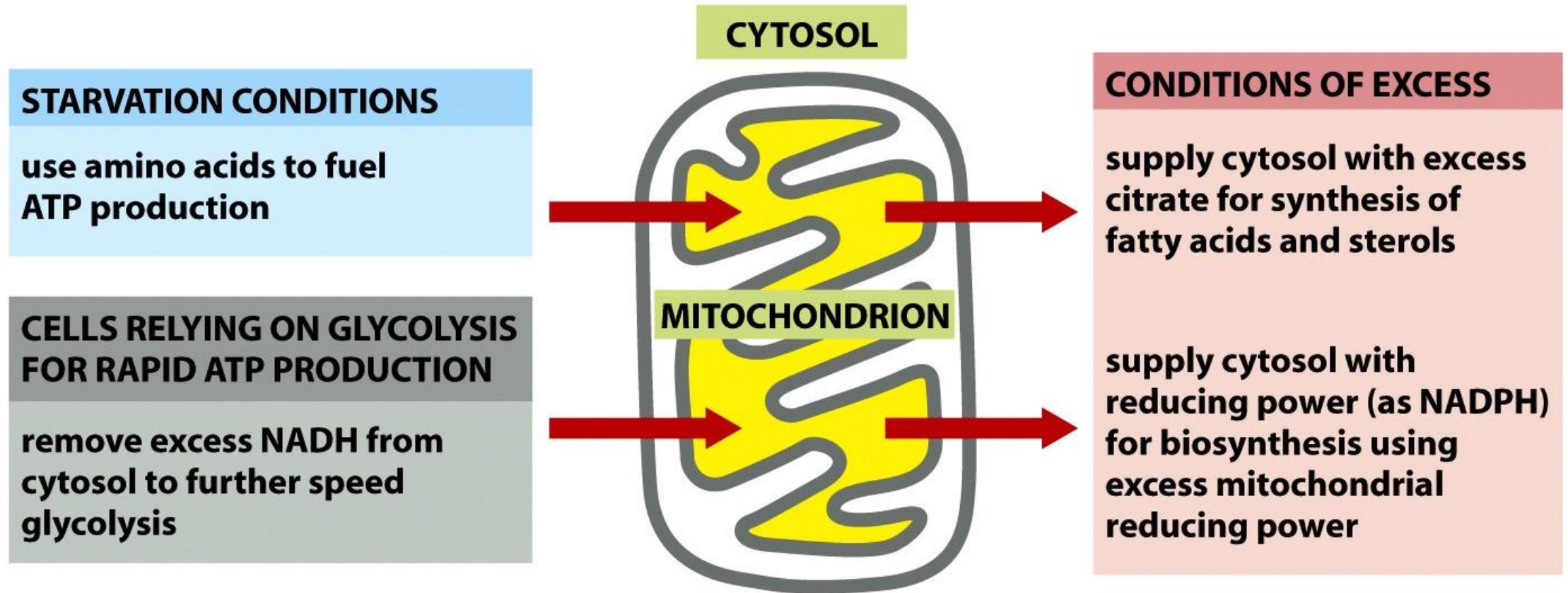
ATP synthase is reversible: the gradient and the ATP/ADP ratio set its direction



KEY POINTS

- **(A) Synthesis** — when the proton-motive force exceeds the ΔG needed for ATP synthesis, H^+ flow into the matrix drives ATP formation.
- **(B) Hydrolysis** — when the gradient collapses (e.g. anoxia, ischaemia) the motor runs backwards, using ATP to pump H^+ out.
- Cells limit this wasteful reversal with the inhibitory protein IF1.

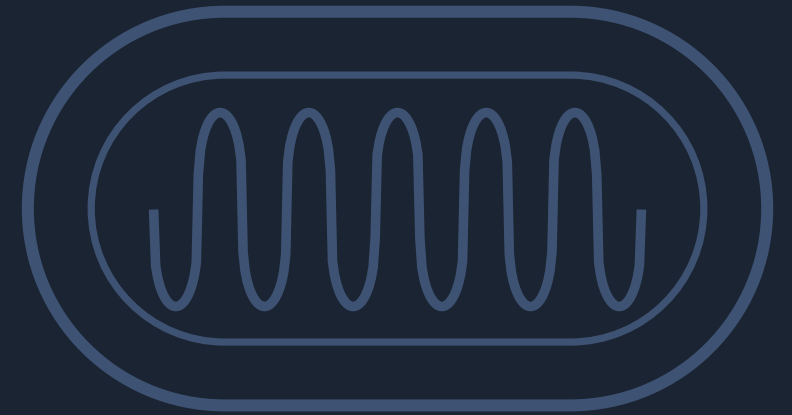
Mitochondria buffer cytosolic metabolism in both starvation and excess



Starvation: mitochondria burn amino acids for ATP. Glycolytic cells: they re-oxidise excess cytosolic NADH, keeping glycolysis running. Excess nutrients: they export citrate for fatty-acid and sterol synthesis, and reducing power (as NADPH) for biosynthesis.

09

A practical case: skeletal muscle



Fibre types, mitochondrial content and Ca^{2+}

Fibre types range from slow oxidative (type I) to fast glycolytic (type IIX)

Muscle Fiber Types

Type 1
(Slow-twitch)



Aerobic (oxidative)
metabolism

Type 2A
(Intermediate fast-twitch)

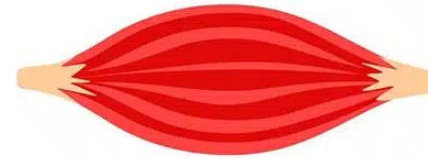


Aerobic (oxidative)
& anaerobic (glycolytic)
metabolism

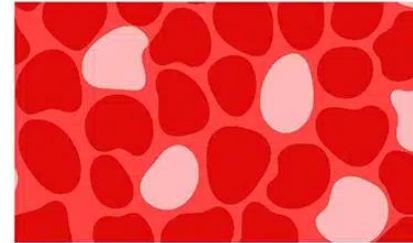
Type Type 2X
(Fast-twitch)



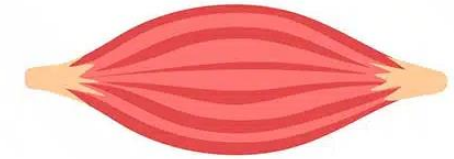
Anaerobic (glycolytic)
metabolism



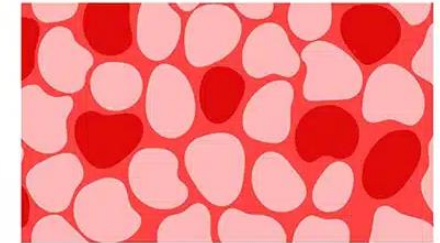
Endurance



Slow twitch muscle fiber



Strength



Fast twitch muscle fiber

Type I

Slow twitch · oxidative · many mitochondria and capillaries · fatigue-resistant (endurance, posture)

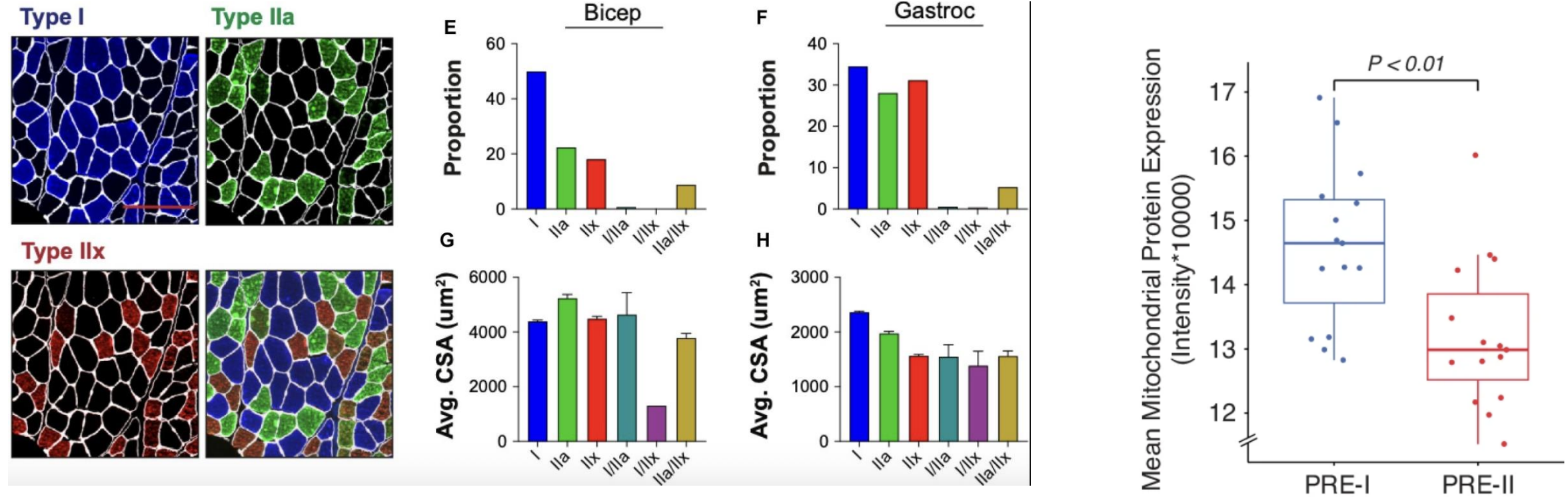
Type IIA

Fast twitch · oxidative–glycolytic · intermediate mitochondrial content

Type IIX

Fast twitch · glycolytic · few mitochondria · high power, fatigues quickly

Mitochondrial content differs between fibre types and muscles



Left: fibre-type immunostaining (type I, IIa, IIx) with fibre-type proportions (E, F) and cross-sectional area, CSA (G, H), in biceps and gastrocnemius. Right: mean mitochondrial protein expression, PRE-I vs PRE-II ($P < 0.01$).

Ca²⁺ links contraction to mitochondrial energy supply

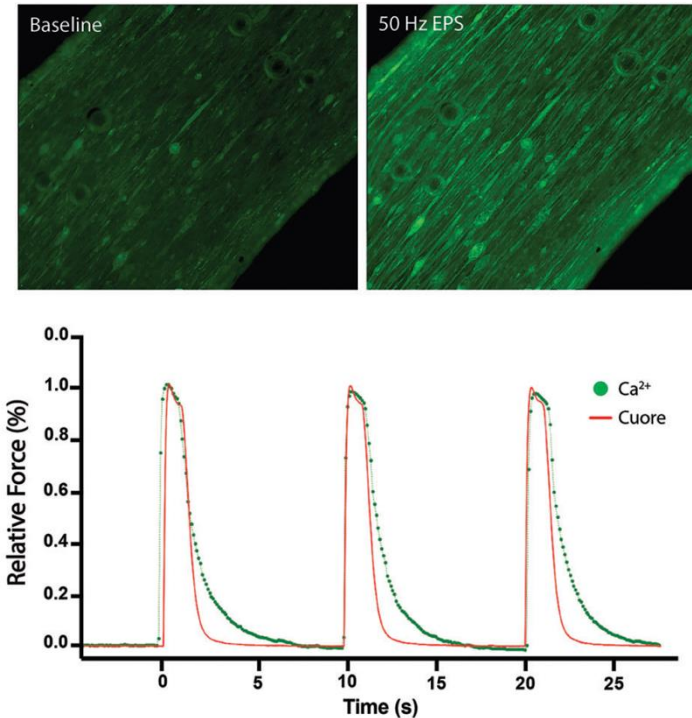


FIGURE LEGEND

- **Left:** Fluo-4 Ca²⁺ imaging during stimulation (movie).
- **Top:** Fluo-4 signal at baseline vs 50 Hz electrical pulse stimulation (EPS).
- **Bottom:** normalised Ca²⁺ transient (green) and force (red) over three tetani.

1 SR releases Ca²⁺
triggers contraction



2 Mitochondria take up Ca²⁺
via the uniporter (MCU);
buffer the cytosolic transient



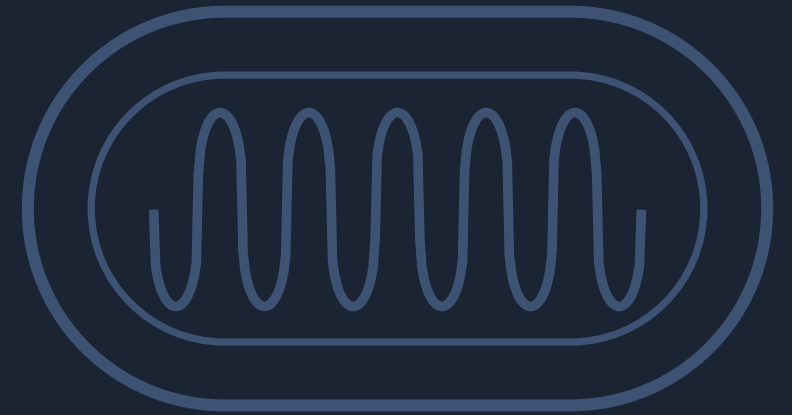
3 Matrix Ca²⁺ activates
dehydrogenases
PDH, isocitrate DH, α-KG DH
→ more NADH



4 ATP supply matches
demand
myosin ATPase and SERCA re-
uptake into the SR

10

What happens when mitochondria fail?



mtDNA and nuclear gene defects · mitochondrial donation

mtDNA disease: maternal inheritance, heteroplasmy and a tissue threshold



16,569 bp

human mtDNA · 37 genes

13 proteins

22 tRNAs

2 rRNAs



Maternal inheritance — mtDNA comes from the oocyte



Heteroplasmy — normal and mutant mtDNA coexist in one cell



Threshold effect — symptoms appear once the mutant fraction exceeds a tissue-specific level



Bottleneck — mutant load can shift sharply from mother to child

13 Protein-coding genes:

- **Complex I (NADH dehydrogenase): 7 subunits** - ND1, ND2, ND3, ND4, ND4L, ND5, ND6
- **Complex IV (Cytochrome c oxidase): 3 subunits** - COX1, COX2, COX3
- **Complex V (ATP synthase): 2 subunits** - ATP6, ATP8
- **Complex II (Succinate dehydrogenase)** – Nuclear genes

Protein-coding genes

- **LHON** — *MT-ND4* m.11778G>A, *MT-ND1* m.3460G>A, *MT-ND6* m.14484T>C · optic nerve
- **NARP / maternally inherited Leigh** — *MT-ATP6* m.8993T>G/C · nerves, retina, brainstem

rRNA genes

- **Aminoglycoside-induced and non-syndromic deafness** — *MT-RNR1* m.1555A>G · cochlea
- Mitochondrial ribosomes resemble bacterial ones — hence the antibiotic sensitivity

tRNA genes

- **MELAS** and **MIDD** — *MT-TL1* m.3243A>G · brain and muscle / pancreas and cochlea
- **MERRF** — *MT-TK* m.8344A>G · muscle, brain

Single large-scale deletions (mostly sporadic)

- **CPEO** → **Kearns–Sayre** → **Pearson**: one clinical spectrum
- Eye muscles, retina, heart conduction; bone marrow and pancreas in Pearson syndrome

Nuclear genes cause most mitochondrial disease, through many pathways

1,136

human mitochondrial proteins (MitoCarta3.0)

13

encoded by mtDNA — all others are nuclear-encoded and imported

Inheritance: autosomal recessive (most) · dominant (e.g. *OPA1*, some *POLG*) · X-linked (e.g. *TIMM8A*)

1 mtDNA maintenance

POLG, TWNK, TK2, MPV17

PEO with multiple deletions · Alpers · depletion syndromes

2 OXPHOS subunits & assembly

NDUFS1, SURF1, SCO2, SDHA

Leigh syndrome · isolated complex deficiencies

3 Fusion & fission

OPA1, MFN2, DNML1

Dominant optic atrophy · CMT2A · fission-defect encephalopathy

4 Protein import

TIMM8A

Mohr–Tranebjærg syndrome (deafness–dystonia)

5 Metabolite carriers (SLC25)

SLC25A4, SLC25A20, SLC25A13

PEO / myopathy · CACT deficiency · citrin deficiency

6 Coenzyme Q₁₀ synthesis

COQ2, COQ4, PDSS2

Primary CoQ₁₀ deficiency (kidney, muscle, CNS)

7 Iron–sulfur clusters

FXN, ISCU, NDU1

Friedreich ataxia · ISCU myopathy · multiple mitochondrial dysfunction

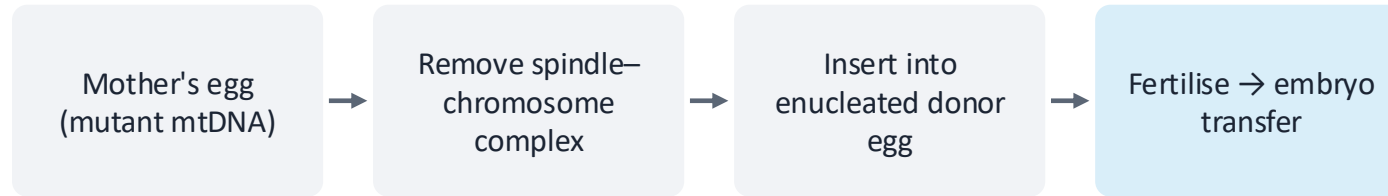
8 Mitochondrial translation

GFM1, TSFM, DARS2

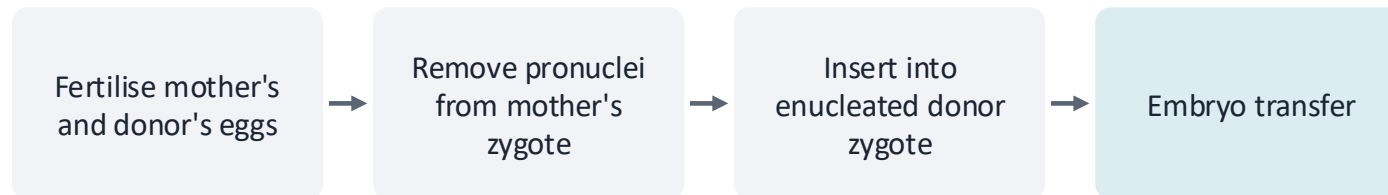
Combined OXPHOS deficiency · LBSL

Mitochondrial donation can prevent transmission of mtDNA disease

Maternal spindle transfer (MST) — before fertilisation



Pronuclear transfer (PNT) — after fertilisation (Newcastle)



Result: nuclear DNA from both parents + mitochondria (mtDNA, 37 genes) from the donor

Open questions: carry-over of maternal mtDNA (low-level heteroplasmy, possible reversion), long-term follow-up, and inheritance of donor mtDNA by future generations through daughters

WHERE DOES IT STAND? (SEPT 2026)

United Kingdom

Legal since 2015; HFEA licenses case by case. Newcastle PNT programme: 8 children born, reported in NEJM (July 2025); maternal mutant mtDNA undetectable or at levels very unlikely to cause disease.

Australia

Legal since 2022 (Maeve's Law). mitoHOPE (Monash): pre-clinical licence Nov 2025, first MST in training Mar 2026; clinical trial now expected from late 2027.

United States

Since Dec 2015, annual appropriations riders bar FDA from accepting MRT applications → clinical use cannot legally proceed.

Take-home messages

1

Chemiosmosis: electron transport builds an H^+ gradient that drives ATP synthesis

2

Fuel oxidation in the matrix (citric acid cycle) loads electrons onto NADH and $FADH_2$

3

Electrons flow $NADH \rightarrow I \rightarrow Q \rightarrow III \rightarrow \text{cyt } c \rightarrow IV \rightarrow O_2$, towards higher redox potential

4

Complexes I, III and IV move H^+ ($\approx 10 H^+$ per NADH); complex II feeds Q without pumping

5

Proton-motive force $\approx 180 \text{ mV} =$ membrane potential (ΔV) + pH gradient (ΔpH)

6

ATP synthase is a reversible rotary motor; the gradient also drives ADP/ATP, P_i and pyruvate transport

7

Uncouplers (protonophores; UCP1 in brown fat) dissipate the gradient: O_2 use continues, ATP synthesis stops, energy becomes heat

8

Muscle: mitochondrial content tracks fibre type; Ca^{2+} couples contraction to ATP supply

9

Disease: mtDNA (maternal, heteroplasmic) or nuclear (Mendelian) defects; mitochondrial donation can prevent mtDNA disease transmission

A1 · Diseases caused by mtDNA mutations (reference)

Gene class	Disease	Gene (common variant)	Main systems affected
Protein-coding	Leber's hereditary optic neuropathy (LHON)	<i>MT-ND4</i> m.11778G>A; <i>MT-ND1</i> m.3460G>A; <i>MT-ND6</i> m.14484T>C	Optic nerve
	Neuropathy, ataxia, retinitis pigmentosa (NARP)	<i>MT-ATP6</i> m.8993T>G	Peripheral nerves, retina, brain
	Maternally inherited Leigh syndrome (MILS)	<i>MT-ATP6</i> m.8993T>G/C; <i>MT-ND</i> genes	Brainstem, basal ganglia
tRNA	MELAS	<i>MT-TL1</i> m.3243A>G (also <i>MT-ND5</i>)	Brain, muscle
	Maternally inherited diabetes and deafness (MIDD)	<i>MT-TL1</i> m.3243A>G	Pancreas, cochlea
	Myoclonic epilepsy with ragged-red fibres (MERRF)	<i>MT-TK</i> m.8344A>G	Muscle, brain
	Mitochondrial cardiomyopathy	tRNA genes (e.g. <i>MT-TI</i>), <i>MT-ND</i> genes	Heart, skeletal muscle
rRNA / tRNA	Non-syndromic and aminoglycoside-induced deafness	<i>MT-RNR1</i> m.1555A>G; <i>MT-TS1</i>	Cochlea
Single large-scale deletion	Chronic progressive external ophthalmoplegia (CPEO)	Single deletion (also <i>MT-TL1</i> point variants)	Extraocular muscles
	Kearns–Sayre syndrome (KSS)	Single deletion	Eye muscles, retina, cardiac conduction
	Pearson syndrome	Single deletion	Bone marrow, exocrine pancreas

A2 · Mitochondrial diseases caused by nuclear gene defects (reference)

Category	Example disease	Key nuclear genes	Main systems affected
mtDNA maintenance	PEO with multiple mtDNA deletions	<i>POLG, POLG2, TWNK, SLC25A4, RRM2B</i>	Extraocular and skeletal muscle
	Alpers–Huttenlocher syndrome	<i>POLG</i>	Brain, liver
	mtDNA depletion syndromes	<i>TK2, DGUOK, SUCLA2, RRM2B, MPV17</i>	Muscle, liver, CNS
OXPHOS subunits & assembly	Leigh syndrome (nuclear)	<i>NDUFS1, NDUFV1, SURF1, COX10, SCO2</i>	Brainstem, basal ganglia
	Complex II deficiency	<i>SDHA, SDHB, SDHAF1</i>	Brain, muscle
	Cytochrome c oxidase (COX) deficiency	<i>SURF1, SCO1, SCO2, COA6, LRPPRC</i>	Brain, muscle, heart, liver
	Complex V deficiency	<i>TMEM70, ATP5F1E</i>	Brain, heart, metabolic crises
Fusion & fission	Dominant optic atrophy	<i>OPA1</i>	Optic nerve
	Charcot–Marie–Tooth type 2A	<i>MFN2</i>	Peripheral nerves
	Encephalopathy due to defective fission	<i>DNM1L, MFF</i>	Brain
Protein import	Mohr–Tranebjærg (deafness–dystonia) syndrome	<i>TIMM8A</i>	Hearing, motor system
Metabolite carriers	Carnitine–acylcarnitine translocase deficiency	<i>SLC25A20</i>	Liver, heart, muscle
	Citrin deficiency (citrullinaemia type II)	<i>SLC25A13</i>	Liver
	Mitochondrial pyruvate carrier deficiency	<i>MPC1</i>	Brain
Coenzyme Q ₁₀ synthesis	Primary CoQ ₁₀ deficiency	<i>COQ2, COQ4, COQ6, PDSS1, PDSS2</i>	Kidney, muscle, CNS
Iron–sulfur clusters	Friedreich's ataxia	<i>FXN</i>	Spinal cord, cerebellum, heart
	ISCU myopathy	<i>ISCU</i>	Skeletal muscle
	Multiple mitochondrial dysfunctions syndromes	<i>NFU1, BOLA3, IBA57, ISCA2</i>	Brain, lung, heart
Mitochondrial translation	Combined OXPHOS deficiency	<i>MRPS16, GFM1, TSFM</i>	Brain, muscle, liver
	mt-aminoacyl-tRNA synthetase disorders (e.g. LBSL, PCH6)	<i>DARS2, RARS2, YARS2, AARS2, VARS2</i>	CNS, muscle

Gene lists are examples, not exhaustive.